CHEMISTRY 2024

(Science Pre-Engineering & Pre-Medical Groups)

TIME: 3 Hours **(85 Marks)**

SECTION 'A' (Multiple Choice Questions)(17)

NOTE: i) This section consists of 17 part questions and all are to be answered. Each question carries one mark.
ii) Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question and its part.
iii) The use of scientific calculator is allowed.

1. Choose the correct answer for each from the given options:

i) Significant figures in 0.0880 are:
* 2 * 3✓ * 4 * 5

ii) Alpha rays consist of:
* Two protons and two electrons
* Two neutrons and two electrons
* Two protons and two neutrons✓
* Two neutrons and one proton

iii) The number of orbitals in fourth energy level is:
* 4 * 9 * 16 ✓ * 32

iv) The number of sigma and pi bonds in C_2H_2 are:
* 2 and 3 * 2 and 2 * 3 and 2 ✓ * 3 and 3

v) This molecule has minimum bond angle:
* CS_2 ✓ * H_2O * NH_3 * BF_3

vi) The geometry of $\overset{+}{NH_4}$ and SO_4^{2-} ions is:
* Tetrahedral ✓
* Trigonal
* Pyramidal
* Square Planer

vii) These pairs do not obey Dalton's Law:
* He and H_2
* H_2 and Ar
* HCl and NH_3^{1-} ✓
* He and NH_3

viii) The vapour pressure of H_2O at 100°C is:
* 76 torr * 5 atms * 101325 Pascal ✓ * 12.5 psi

ix) London dispersion force is stronger in:
* F_2 * Cl_2 * Br_2 * I_2 ✓

x) Diamond is an example of:
* Metallic Solid
* Molecular Solid
* Ionic Solid
* Covalent Solid ✓

xi) If $a \neq b \neq c$ and $\alpha = \gamma = 90^\circ$ but $\beta \neq 90^\circ$, then the cry structure is:
* Orthorhombic
* Hexagonal
* Monoclinic✓
* Triclinic

xii) If a catalyst is added in a chemical system at equilibrium, the v of K_c:
* will decrease
* will increase
* will become zero
* will not be changed✓

xiii) This oxide is amphoteric:
* K_2O
* CO_2
* CaO
* Al_2O_3✓

xiv) If the rate law of reaction is R = K, its order of reaction is:
* Zero
* first
* second
* third

xv) The oxidation numbers of N in NH_4NO_3 are:
* $-3, +5$ ✓
* $-3, +3$
* $-5, +5$
* $-1, +1$

xvi) This compound gives acidic solution when dissolved in water:
* NaCl
* NH_4Cl ✓
* CH_3COONH_4
* Na_2SO_4

xvii) At pH = 0, the universal indicator shows this colour:
* Red✓
* Orange
* Yellow
* Green

Atomic Masses:

(H = 1amu, C = 12 amu, Al = 27amu, O = 16amu
Cl = 35.5 amu, S = 32amu, Na = 23 amu, Mg = 24a

SECTION 'B' (Short-Answer Questions) (36)

NOTE: Answer any NINE part questions. All questions c equal marks.

2. i) Define any four of the following:
* Stoichiometry
* Exponential notation
* Unit Cell
* Latent Heat of Fusion
* Energy of Activation

ANSWER:

STOICHIOMETRY

Stoichiometry deals with the study of the quantitative relationship between reactants and products in a chemical reaction by using a balanced chemical equation.

EXPONENTIAL NOTATIONS

Exponential Notation is the way of expressing very large or very s numbers in a compact form that is easy to use in computations.

UNIT CELL

"The smallest block of a crystal which consists of atoms, ions or m and has all characteristic of that substance crystal and repeat block in space produces a crystal lattice"

LATENT HEAT OF FUSION

The amount of heat required to convert a unit mass or 1g of a solid at its melting point into a liquid without an increase in temperature is called latent heat of fusion.

ENERGY OF ACTIVATION

The least amount of energy required to convert stable molecule into active molecule is called energy of activation".

(i) Chlorine gas is produced on an industrial scale by electrolysis of a aqueous solution of $NaCl$

$$2NaCl + 2H_2O \rightarrow Cl_2 + H_2 + 2NaOH$$

Calculate:
- The volume of Cl_2 at S.T.P produced from 58.5 g of NaCl.
- The total number of molecules of Cl_2 produced.

ANSWER:

DATA:

GIVEN:

$Given\ Mass\ of\ NaCl = 58.5g$

$Molecular\ mass\ of\ NaCl = 23 + 35.5 = 58.5$

REQUIRED:

Volume of Cl_2 =?

number of molecules of Cl_2 =?

SOLUTION:

$$\because Number\ of\ moles = \frac{Given\ Mass}{Molecular\ mass}$$

$$Number\ of\ moles\ of\ NaCl = \frac{58.5}{58.5} = 1\ mole$$

According to the given equation

$2\ moles\ of\ NaCl\ gives\ 1\ mole\ of\ Cl_2$

$1\ mole\ of\ NaCl\ gives\ \frac{1}{2}\ mole\ of\ Cl_2$

VOLUME OF Cl_2 AT S.T.P

$1\ mole\ of\ Cl_2\ gas\ occupied = 22.4\ dm^3$

$\frac{1}{2}\ mole\ of\ Cl_2\ gas\ occupied = 22.4\ dm^3 \times \frac{1}{2}$

$\frac{1}{2}\ mole\ of\ Cl_2\ gas\ occupied = 11.2\ dm^3$

TOTAL NUMBER OF MOLECULES OF Cl_2 PRODUCED

We know that

$1\ mole\ of\ Cl_2\ gas\ contains = 6.0225 \times 10^{23}\ molecules$

$\frac{1}{2}\ mole\ of\ Cl_2\ gas\ contains = 6.0225 \times 10^{23}\ molecules \times \frac{1}{2}$

$\frac{1}{2}$ *mole of* Cl_2 *gas contains* $= 3.01 \times 10^{23}$ *molecules*

iii) 53.5g of Aluminium is burnt with 22.4g of oxygen to given bri white flame of Al_2O_2.

$$4Al + 3O_2 \rightarrow 2Al_2O_3$$

Which reactant is limiting reactant? Also find the mass of A produced.

ANSWER:

DATA:

GIVEN:

$Given\ Mass\ of$ Aluminium $= 53.5g$
$Given\ Mass\ of\ oxygen = 22.4g$
$Atomic\ mass\ of\ Aluminum = 27amu$
$Molecular\ mass\ of\ Oxygen = 16 \times 2 = 32amu$
$Molecular\ mass\ of\ Al_2O_3 = 27 \times 2 + 16 \times 3 = 102amu$

REQUIRED:

$Mass\ of\ Al_2O_3 = ?$

SOLUTION:

$$\because Number\ of\ moles = \frac{Given\ Mass}{Molar\ mass}$$

$$Number\ of\ moles\ of\ Al = \frac{53.5}{27} = 1.98\ mole$$

$$Number\ of\ moles\ of\ O_2 = \frac{22.4}{32} = 0.7\ mole$$

According to the given equation

$4\ moles\ of\ Al\ gives = 2\ mole\ of\ Al_2O_3$

$1\ moles\ of\ Al\ gives = \frac{2}{4}\ mole\ of\ Al_2O_3$

$1.98\ moles\ of\ Al\ gives = \frac{1}{2} \times 1.98\ mole\ of\ Al_2O_3$

$1.98\ moles\ of\ Al\ gives = 0.99\ mole\ of\ Al_2O_3$

Also

$3\ moles\ of\ O_2\ gives = 2\ mole\ of\ Al_2O_3$

$1\ moles\ of\ O_2\ gives = \frac{2}{3}\ mole\ of\ Al_2O_3$

$0.7\ moles\ of\ O_2\ gives = \frac{2}{3} \times 0.7\ mole\ of\ Al_2O_3$

$0.7\ moles\ of\ O_2\ gives = 0.46\ mole\ of\ Al_2O_3$

Since Oxygen gives least amount of product, therefore it is a limiting reactant

$$Mass\ of\ Al_2O_3 = Molecular\ mass \times No. of\ moles$$

$$Mass\ of\ Al_2O_3 = 102 \times 0.46$$

$$Mass\ of\ Al_2O_3 = 46.92\ g$$

State 'Hund's rule'. Write electronic configuration of the following:
r(Z = 24) $^*Sr^{++}(Z = 38)$ $^*S^{++}(Z = 16)$

ANSWER:

HUND'S RULE:

This principle state that, "In available degenerated orbitals (p, d and f), electrons are distributed in such a way that the maximum number of half-filled orbitals (single electron in orbitals) are obtained".

ELEMENTS	CONFIGURATION
Cr(Z = 24)	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1, 3d^5$
Sr^{++}(Z = 38)	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}, 4p^6$
S^{--}(Z = 16)	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6$

Give any four scientific reasons for the following:

Evaporation is a cooling process.

ANSWER: Evaporation is an endothermic process. When it starts, the high-energy molecules leave the liquid surface. Therefore the average kinetic energy of the remaining molecules decreases. As a consequence, the temperature of the liquid falls. Therefore evaporation is a cooling process

Graphite conduct electricity but diamond does not.

ANSWER: In graphite carbon atom is sp^2 hybridized. Each carbon atom uses three electrons in bond formation while 4th valence electron of each carbon atom is free to move which is responsible for the flow of electric current. But in diamonds no free electron available for conducting electricity

Boiling point of H_2O is higher than HF.

ANSWER: Fluorine can form one hydrogen bond with partial positive hydrogen of neighbouring molecules. While water forms two hydrogen bonds per molecule, the reason is that water has two partial positive hydrogen atoms and two lone pairs on oxygen atoms. Hence water has strong hydrogen bonding than HF which leads to its high boiling point.

Pressure cooker is used for rapid cooking.

ANSWER: The pressure cooker is a closed container the vapour of water formed is not allowed to escape so the vapour of water develop more pressure in the cooker so boiling temperature increases, so more heat is absorbed in water and food cooked quickly.

Milk sour more rapidly in summer than in winter.

ANSWER: Souring of milk is a biochemical process which is fast at high temperatures. In summer temperature is higher than in winter, therefore milk gets sour more rapidly in summer than in winter.

vi) A $100cm^3$ sample of hydrogen effuse four times as rapidly as ... of an unknown gas. Calculate the molecular mass of the unknown ...

ANSWER:

DATA:

GIVEN:

$Volume\ of\ Hydrogen\ gas = V_1 = 100cm^3$

$Volume\ of\ unknown\ gas = V_2 = 100cm^3$

$Molecular\ mass\ of\ Hydrogen = M_1 = 2amu$

$Rate\ of\ ef\ effusion\ of\ Hydrogen = r_1 = 4r$

$Rate\ of\ ef\ effusion\ of\ unknown = r_2 = r$

REQUIRED:

$Molecular\ mass\ of\ unknown\ gas = M_2 =$

SOLUTION:

According to Graham's Law of Diffusion

$$\because \frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$$

$$\frac{4r}{r} = \sqrt{\frac{M_2}{2}}$$

$$4 = \sqrt{\frac{M_2}{2}}$$

Squaring on both sides

$$16 = \frac{M_2}{2}$$

$$M_2 = 32\ amu$$

vii) Drive Van der Waal's equation for real gases by correcting vol... and pressure.

ANSWER:

VOLUME CORRECTION

When pressure is applied to a gas, the molecules come closer to ... other. By continuous increase in pressure a point is reached ... molecules cannot further be compressed since repulsive forces ... created. This indicates that gas molecules have definite volume. Alth... this is very small but not negligible. Keeping in view the definite volum... gas molecules, van der Waal calculated the actual volume of gas ... follows:

$$V = V_{vessel} - nb - - - - - (i)$$

Where,

V = free volume

V_{vessel} = Volume of the gas molecules are present.
n = number of moles
b = excluded volume of gas molecules per mole in highly compressed gaseous sate. It is theoretical volume.
The value of "b" depends upon the size of gas molecules. Here it should be noted that 'b' is not actual volume of gas but it is assumed four times the actual volume of molecules.

$$b = 4V_{vessel}$$

Where, V_{vessel} is actual volume of one mole of a gas.

PRESSURE CORRECTION

Van der Waal corrected the pressure produced by the molecules of the real gases.

Consider a molecules "A" in the interior of the gas, which is completely surrounded and attracted by the other gas molecules "B". The resultant attractive force on the molecules "A" due to all the surrounding "B" molecules is zero as shown in figure. However, as this molecule "A" approaches to strike the wall of vessel, it experience a net inward pull due to the attractive forces of molecules of "B". It means that pressure produced on the wall would be little bit lesser than pressure of an ideal gas molecule.

Therefore;

$$P_{observed} = P_{ideal} - P_{less}$$

If observed pressure is simply indicated by P, ideal pressure P_i and less pressure P_L

Then equation will be:

$$P = P_i - P_L$$

$$P_i = P + P_L - - - - (i)$$

Since P_L is the pressure drop due to backward pull of striking molecules, it depends upon number of particles (A and B) per unit volume.

$$P_L \propto [A].[B]$$

$$P_L = a[A].[B]$$

Where "a" is the constant of proportionality

The concentration of "A" and "B" is equal to number of moles divided by volume, therefore

$$P_L = a\left(\frac{n}{V}\right)\left(\frac{n}{V}\right)$$

$$P_L = \frac{an^2}{V^2}$$

On substituting the value of "P_L" in equation (i) we get

$$P_i = P + \frac{an^2}{V^2}$$

$$\because PV = nRT$$

The corrected pressure and volume is now put in general gas equation, we get

$$\left(P + \frac{an^2}{V^2}\right)(V_{vessel} - nb) = nRT$$

$\because V_{vessel} =$

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

If $n = 1$, then above equation will be

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

This is van der Waal's equation. Here 'a' and 'b' are van der Waal's constant.

viii) The solubility of $Mg(OH)_2$ at 25°C is $4.6 \times 10^{-3} g/100cm^3$ calculate the solubility product of the $Mg(OH)_2$.

ANSWER:

DATA:

GIVEN:

Solubility of $Mg(OH)_2 = 4.6 \times 10^{-3} g/100cm^3$

Molecular Mass of $Mg(OH)_2 = (24) + 2\,(16 + 1) = 24 + 34 =$ 58amu

REQUIRED:

Solubility product of $Mg(OH)_2 = ?$

SOLUTION:

$$Concentration = \frac{solubility}{Molar\ mass}$$

$$Conc.\,of\ Mg(OH)_2 = \frac{0.0046}{58} = 7.9 \times 10^{-5} mole/cm^3$$

$$Concentration\ of\ Mg^+ = 7.9 \times 10^{-5} mole/cm^3$$

$$Concentration\ of\ OH^- = 2(7.9 \times 10^{-5} mole/cm^3)$$

$$= 1.58 \times 10^{-4}\ mole/cm^3$$

$$\because K_{sp} = [Mg^+][OH^-]^2$$

$$K_{sp} = (7.9 \times 10^{-5})(1.58 \times 10^{-4})^2$$

$$K_{sp} = (7.9 \times 10^{-4})(2.496 \times 10^{-8})$$

$$K_{sp} = 1.972 \times 10^{-11} mole^3/cm^9$$

ix) Explain Bronsted Lowry Theory of acids and bases. Also explain conjugate acid-base pair.

ANSWER:

BRONSTED-LOWRY THEORY OF ACIDS AND BASES

According to this theory

...cid is a species which tends to, donate a proton (protogenic) whereas ...base is a species that accepts a proton (protophillic). Further, an acid-...se reaction is the transfer of a proton from the acid to base".

...ONSTED-LOWRY ACID:

"...substance donates protons (H+) during chemical reactions is called ...d".

OR

"...n acid is any molecule or ions that can proton (H+ ion) donate in a ...emical reaction".

...ONSTED-LOWRY BASE:

"...substance accepts protons (H^+) during chemical reactions is called ...se".

OR

"...base is any molecule or ions that can proton (H+ ion) accept in a ...mical reaction".

...PLANATION

...light of definition, all hydrogen-containing substances (either ...ecules or ions) which are capable of giving up a proton (HCl, HNO_3, ...SO_4, H_2O, H_3O^+, NH_4^+ etc) are gathered in the list of acids whereas all ...e substances which have the ability to accept a proton (NH_3, H_2O, ... Cl^-, CO_3^{-2}, HSO_3^-) are categorized as bases.

...NJUGATE ACID BASE PAIRS

...onjugated acid base pair represent two species, one on the left side ...other on the right side of equation with the difference of one proton. ...mportant to note that strong acid has weak conjugate base and weak ... has strong conjugate acid.

$$\underset{\text{(Bronsted acid)}}{H_2S_{(g)}} + \underset{\text{(Bronsted base)}}{NH_{3(aq)}} \leftrightharpoons \underset{\text{(conjugate acid)}}{HS^-_{(aq)}} + \underset{\text{(conjugate base)}}{NH^+_{4(aq)}}$$

...e reaction $2NO + Cl_2 \rightarrow 2NOCl$ was studied at 25°C the following ...s were obtained:

Exp. No	Initial Concentration mol/dm³		Initial rate mol/dm³.sec
	[NO]	[Cl_2]	
1	0.1	0.1	2.52×10^{-3}
2	0.1	0.2	5.04×10^{-3}
3	0.2	0.1	10.05×10^{-3}

...the basis of above information illustrate the Rate law and find the ...f reaction.

...TION:

(i) When the initial concentration of NO is doubled (experiment 3), initial rate is increased by 2 times. Thus rate law with respect to A is

$$R \propto [NO]^2$$

(ii) When the initial concentration of Cl_2 is doubled (experiment 2), initial rate is also doubled. Thus rate law with respect to A is.

$$R \propto [Cl_2]^1$$

(iii) The overall rate law now written as

$$R \propto [NO]^2[Cl_2]^1$$
$$R = K[NO]^2[Cl_2]^1$$

(iv) Order of reaction is thus $2 + 1 = 3$

xi) Define Colligative properties. Explain elevation of boiling point and depression of freezing point.

ANSWER:

COLLIGATIVE PROPERTIES

Certain physical properties of solutions depend upon the number of solute particles rather than their chemical nature, these are referred as colligative properties. (Colligative means collective). Examples the lowering of vapour pressure, the elevation of boiling point, the depression of freezing point and osmotic pressure. We specifically focus on the colligative properties of solutions of non-volatile and electrolyte solutes.

ELEVATION OF BOILING POINT

The temperature at which the vapour pressure of a liquid becomes equal to atmospheric pressure (1 atm) is known as the normal boiling point. If a non-volatile and non-electrolyte solute (solid) is dissolved into a volatile solvent, its vapour pressure is found to be lowered 1 atmosphere. Since liquid boil only when its vapours pressure becomes equal to atmospheric pressure, some more heat is needed the solution to reach its vapours pressure 1 atmosphere. This concluded that the boiling point of the solution is greater than the boiling of the pure solvent. "Boiling point of a solution of non-volatile solute is higher than pure solvent", this is a colligative property known as elevation of boiling point (ΔT_b)"

DEPRESSION OF FREEZING POINT

We know that the kinetic energy of liquid molecules decreases on cooling and molecules approach one another more closely and the liquid finally turns into a solid. "Freezing point of a liquid is the temperature at which liquid and solid phases co-exist in equilibrium with the same vapours pressure".

...presence of a non-volatile solute lowers the freezing point of the solvent. These phenomena can be explained in terms of vapour pressure. Since at freezing point, the vapour pressure of both liquid and solid phases must be the same, the addition of solute causes a fall in vapour pressure. Thus in order to re-establish the equilibrium of the two phases, the temperature of the solution is consequently lowered below the freezing point of the pure solvent.

Calculate the Heat of formation of Fe_2O_3 at 25°C:

$Fe + \frac{3}{2}O_2 \rightarrow Fe_2O_3$ $\Delta H_f^0 = ?$

$H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ $\Delta H = -286\ KJ/mole$

$Fe_2O_3 + 3H_2O \rightarrow 2Fe(OH)_3$ $\Delta H = +289\ KJ/mole$

$Fe + 3H_2O \rightarrow Fe(OH)_2 + \frac{3}{2}H_2$ $\Delta H = +161\ KJ/mole$

SOLUTION:

$H_2 + \frac{1}{2}O_2 \longrightarrow H_2O$ $\Delta H = -286KJ/mole$

$Fe_2O_3 + 3H_2O \longrightarrow 2Fe(OH)_3$ $\Delta H = +289J/mole$

$Fe + 3H_2O \longrightarrow Fe(OH)_3 + \frac{3}{2}H_2$ $\Delta H = -161J/mole$

Equation (1) multiply by 3 then subtracts Equation (2) from equation (1).

$3H_2 + \frac{3}{2}O_2 \longrightarrow 3H_2O$ $\Delta H = -858KJ/mole$

$Fe_2O_3 + 3H_2O \longrightarrow 2Fe(OH)_3$ $\Delta H = +289J/mole$

$-\quad -\quad -$

$3H_2 + \frac{3}{2}O_2 + Fe_2O_3 + 2Fe(OH)_3 \longrightarrow 6H_2O$

$\Delta H = -1147J/mole$

Equation (3) multiply by 2 then Subtract Equation (3) from equation (4)

$3H_2 + \frac{3}{2}O_2 + Fe_2O_3 + 2Fe(OH)_3 \longrightarrow 6H_2O$

$\Delta H = -1147J/mole$

$2Fe + 6H_2O \longrightarrow 2Fe(OH)_3 + 3H_2$

$\Delta H = -322J/mole$

$-\quad -\quad -\quad +$

$2Fe - \frac{3}{2}O_2 \longrightarrow Fe_2O_3$ $\Delta H_f = -825J/mole$

Differentiate any two of the following:

VBT and MOT.

Line Spectrum and Continuous Spectrum

Ideal Solution and Non-ideal Solution

ANSWER:

VBT AND MOT

VALENCE BOND THEORY	MOLECULE ORBITAL THEOR
Valence bond theory tells that only some valence electrons are involved in the bond formation	Molecular orbital theory tells that bond formation occurs due to the involvement of all the valence electrons of interacting atoms.
According to valence bond theory, both the concerned atomic orbitals possess their individual identify	According to the molecular orbital theory, both the concerned atomic orbitals donot possess their individual identify.
VBT doesn't introduce any adequate idea about the bond order.	MOT given adequate idea about the bond order by virtue of which, the bond can be identified as single, double and triple bond.
The paramagnetic behavior of molecules such as oxygen molecule O_2 cannot be explained by valence bond theory.	The paramagnetic behavior of molecules are very well explained using molecular orbital theory.

LINE SPECTRUM AND CONTINUOUS SPECTRUM

CONTINUOUS SPECTRUM	LINE SPECTRUM
It consists of a band of seven colours without border or separation lines	It consists of a band of seven colours with border or separation lines.
colours of continuous spectrum are diffused into each other	colours of continuous spectrum are not diffused into each other
It is produced when light is coming from the sun or incandescent lamps.	It is produced when atom or substance stimulated by heat or electric current.

IDEAL SOLUTION AND NON-IDEAL SOLUTION

IDEAL SOLUTION	NON-IDEAL SOLUTION
The solution which perfectly obeys Raoult's law is referred as ideal solutions.	The solution that does not perfectly obey Raoult's law is referred as non-ideal solution.
It obeys Raoult's law at all temperatures and concentrations.	It does not obey Raoult's law at all temperature and concentration.
No heat absorbed or evolved in the formation of the ideal	Heat is absorbed or evolved in the formation of the non-ideal

solution.	solution.
No change in the volume takes place in the formation of the ideal solution.	The change in the volume may take place in the formation of a non-ideal solution.

xiv) At 30°C the vapour pressure of water is 31.6 torr. If 30g of urea (molecular weight = 60) are dissolved in 180g of water. Calculate the lowering of vapour pressure.

ANSWER:

DATA:

GIVEN:

$Vapour\ pressure\ of\ water = P^o = 31.6\ torr$

$Given\ mass\ of\ Urea = 30g$

$Given\ mass\ of\ Water = 180g$

$Molecular\ weight\ of\ water = 18amu$

$Molecular\ weight\ of\ Urea = 60amu$

REQUIRED:

$Lowering\ of\ vapour\ pressure = \Delta P = ?$

SOLUTION:

$$\because Number\ of\ moles = \frac{Given\ Mass}{Molar\ mass}$$

$$Number\ of\ moles\ of\ Water = n_1 = \frac{180}{18} = 10\ moles$$

$$Number\ of\ moles\ of\ Urea = n_2 = \frac{30}{60} = 0.5\ moles$$

First we determine the mole fraction of glucose.

$$X_2 = \frac{n_2}{n_1 + n_2}$$

$$X_2 = \frac{0.5}{180 + 0.5} = 0.00278$$

Now applying second form of Raoult law.

$$\because \Delta P = P^o X_2$$

$$\Delta P = (31.6)(0.00278)$$

$$\Delta P = 0.088\ torr$$

SECTION "C" (Detailed Answer Questions)(32)

NOTE: Attempt any Two questions from this section. All questions carry equal marks.

3. a) Write the postulates of Bohr's atomic model. Starting from $\Delta E = E_2 - E_1$ derive an expression for the wave number of radiations emitted by an electron.

ANSWER:

POSTULATE OF BOHR'S ATOMIC THEORY

Postulates of Bohr's theory are given below:

(i) Electrons revolves around the nucleus in circular orbit situated a distance from nucleus with definite energy.

(ii) As long as an electrons remains in an appropriate Orbit, it neither loses, nor gains energy. Hence each Orbit has a fixed energy level, however the energy of orbit increases with the increase distance from nuclei.

(iii) During excitations, electrons of absorbs some quantized energy jump from a lower energy orbit to an appropriate higher energy orbit, when it return back to lower energy orbit it emits quantized energy.

$$\Delta E = E_2 - E_1 = h\upsilon$$

Here "υ" is frequency of radiation and "h" is plank constant

(iv) Electron can move only in those orbit in which the angular momentum of electrons (mvr) is integral multiple of nh/2π

$$mvr = \frac{nh}{2\pi}$$

Here 'm' and 'v' are the mass of velocity of electron and 'r' is the radius of orbit.

DERIVATION:

When an electron jumps from higher energy to lower energy orbit definite amount of energy is emitted, which is equal to the difference energy among both energy levels, i.e.

$$\Delta E = E_2 - E_1 ----(i)$$

$$\because E_1 = -\frac{me^4}{8\varepsilon_o^2 h^2}\left(\frac{1}{n_1^2}\right), \quad E_2 = -\frac{me^4}{8\varepsilon_o^2 h^2}\left(\frac{1}{n_2^2}\right)$$

On substituting the values of "E_1" and "E_2" in equation (i), we get

$$\Delta E = \left\{-\frac{me^4}{8\varepsilon_0^2 h^2}\left(\frac{1}{n_2^2}\right)\right\} - \left\{-\frac{me^4}{8\varepsilon_o^2 h^2}\left(\frac{1}{n_1^2}\right)\right\}$$

$$\Delta E = \frac{me^4}{8\varepsilon_o^2 h^2}\left(-\frac{1}{n_2^2} + \frac{1}{n_1^2}\right)$$

$$\Delta E = \frac{me^4}{8\varepsilon_0^2 h^2}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

According to plank's quantum theory

$$\because \Delta E = h\nu$$

$$\therefore h\nu = \frac{me^4}{8\varepsilon_0^2 h^2}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$v = \frac{me^4}{8\varepsilon_o^2 h^3}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) Hertz$$

Where "v" is the frequency
The wave number of absorbed or emitted photons is

$$v = \bar{V}C$$

Or

$$\frac{me^4}{8\varepsilon_o^2 h^3}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) = \bar{V}C$$

$$\bar{V} = \frac{me^4}{8\varepsilon_o^2 h^3 C}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\because R_H = \frac{me^4}{8\varepsilon_o^2 h^3 C}$$

$$\bar{V} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \text{ Drived}$$

Where R_H is Rydberg constant and its value is $1.09678 cm^{-1}$

b) Define Corrosion. Balance the following equations by ion-electronic method:

- $Cr(OH)_3 + SO_4^{2-} \rightarrow CrO_4^{2-} + SO_3^{2-}$ (Basic medium)
- $Cr_2O_7^{2-} + I_2 \rightarrow Cr^{3+} + IO_3^-$ (Acidic medium)

ANSWER:

CORROSION:

"A spontaneous process which surface atoms of metals get oxidized (harmful oxides) due to the action of surrounding medium is known as corrosion".

SOLUTION:

$$Cr(OH)_3 \longrightarrow CrO_4^{2-} \quad \text{Oxidation } +3 \text{ to } +6$$

$$SO_4^{2-} \longrightarrow SO_3^{2-} \quad \text{Reduction } +6 \text{ to } +4$$

$$Cr(OH)_3 + 5OH^- \longrightarrow CrO_4^{2-} + 4H_2O$$

$$SO_4^{2-} + H_2O \longrightarrow SO_3^{2-} + 2OH^-$$

$$Cr(OH)_3 + 5OH^- \longrightarrow CrO_4^{2-} + 4H_2O + 3e$$

$$SO_4^{2-} + H_2O + 2e \longrightarrow SO_3^{2-} + 2OH^-$$

$$[Cr(OH)_3 + 5OH^- \longrightarrow CrO_4^{2-} + 4H_2O + 3e] \times 2$$

$$[SO_4^{2-} + H_2O + 2e \longrightarrow SO_3^{2-} + 2OH^-] \times 3$$

$$2Cr(OH)_3 + 10OH^- \longrightarrow 2CrO_4^{2-} + 8H_2O + 6e$$

$$3SO_4^{2-} + 3H_2O + 6e \longrightarrow 3SO_3^{2-} + 6OH^-$$

$$2Cr(OH)_3 + +4OH^- + 3SO_4^{2-} \longrightarrow 2CrO_4^{2-} + 3SO_3^{2-} + 5H_2O$$

* $Cr_2O_7^{2-} + I_2 \rightarrow Cr^{3+} + IO_3^-$ (Acidic medium)

SOLUTION:

$I^- \longrightarrow IO_3^{-1}$ Oxidation -1 to +5

$Cr_2O^{2-}_7 \longrightarrow Cr^{3+}$ Reduction +6 to +3

$I^- + 3H_2O \longrightarrow IO_3^{-1} + 6H^+$

$Cr_2O^{2-}_7 + 14H^+ \longrightarrow 2Cr^{3+} + 7H_2O$

$I^- + 3H_2O \longrightarrow IO_3^{-1} + 6H^+ + 6e$

$Cr_2O^{2-}_7 + 14H^+ + 6e \longrightarrow 2Cr^{3+} + 7H_2O$

Adding both equations

$I^- + 3H_2O \longrightarrow IO_3^{-1} + 6H^+ + 6e$

$Cr_2O^{2-}_7 + 14H^+ + 6e \longrightarrow 2Cr^{3+} + 7H_2O$

$I^- + Cr_2O^{2-}_7 + 8H^+ \longrightarrow IO_3^{-1} + 2Cr^{3} + 4H_2O$

4. a) Write the postulates of VSEPR model. State the shapes of following molecules with the help of hybridization

*H_2O * BF_3

ANSWER:

VALENCE SHELL ELECTRON PAIR REPULSION THEORY (VSEPR)

This approach to the structure of a covalent molecular is due to sidgwick and powell (1940). They pointed out that the shapes of molecules can be determined by the repulsion between the electron pairs present in valence shell of central atom.

POSTULATES OF VSEPR

The main postulates of this theory are:

1. There maybe two types of electron pairs surrounding the central atom

(a) BOND PAIRS: These are the result of the sharing of unpaired electrons of central atom with unpaired electron of surrounding atom. These are also called active set of electrons.

(b) LONE PAIRS: These are the paired electron or central atom which do not take part in sharing. They are also called non-bonding pairs. They are also considered to be active set of electrons.

2. Being similarly charge (i.e. negative) the bond pairs as well as the lone pairs repel each other.

3. Due to repulsion, the electron pairs of central atom try to be as far apart as possible; hence they orient themselves in space in such a manner that force of repulsion between them is minimized.

3. The force of repulsion between lone pairs and bond pairs is not the same. The order of repulsion is as follow.

lone pair – Lone pair > Lone pair – Bond pair > Bond pair – Bond pair.

5. In case of molecules with double and triple bonds, the π electron pairs are not considered to be an active set of electrons, hence not included in the count of total electron pairs.
6. The shape of molecules depends upon total number of electron pairs (bonding and lone pairs).

SHAPE OF H_2O

Lewis structure of H_2O molecules is drawn as [redacted]
Electronic information around the centre atom is given as
No. of bond pair electrons = 2
No. of Lone pair electrons = 2
The repulsion of two lone pair electrons reduces the bond angle from 109.5^o to 104.5^o
Hence the shape of H_2O molecule is angular

SHAPE OF BF_3

Lewis structure of BF_3 molecules is drawn as [redacted]
Electronic information around the centre atom is given as
No. of bond pair electrons = 3
No. of Lone pair electrons = 3
The repulsion of three lone pair electrons and three bond pairs of elector distribute bond angle equally i.e. 120^o.
Hence the shape of BF_3 molecule is Trigonal Planar.

b) State law of mass action. Derive the expression of K_c for the general reversible reaction $aA + bB \leftrightharpoons cC + dD$ and also give the relationship between K_p, K_c and K_n.

ANSWER:

LAW OF MASS ACTION

"The rate at which a substance reacts is proportional to its active mass and the rate of a chemical reaction is proportional to the product of the active masses of reactants".

OR

"The rate of a chemical reaction is directly proportional to the product of the molar concentrations of the reactants at a constant temperature, at any given time".

EXPRESSION OF A GENERAL REVERSIBLE REACTION

Consider the following hypothetical chemical reaction,

$$aA + bB \rightleftharpoons cC + dD$$

Where a, b, c and d are the number of moles of A, B, C and D.
According to law of mass action,

$$Rate\ of\ forward\ reaction \propto [A]^a[B]^b$$
$$= K_f[A]^a[B]^b$$
$$Rate\ of\ reverse\ reaction \propto [C]^c[D]^d$$
$$= K_r[C]^c[D]^d$$

At equilibrium rate of forward reaction is equal to rate of reverse rea
therefore

$$Rate\ of\ forward\ reaction = Rate\ of\ reverse\ reaction$$
$$K_f[A]^a[B]^b = K_r[C]^c[D]^d$$
$$\frac{K_f}{K_r} = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

$\because \frac{K_f}{K_r}$

$$K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

RELATIONSHIP BETWEEN K_p, K_c AND K_n

For a homogenous gaseous reaction in which all reacting species appe the gaseous state, the concentration of products and reactants can al expressed in term of their partial pressure because it is easier for a g measure its pressure rather than concentration. For the reaction

$$aA_{(g)}bB_{(g)} \leftrightharpoons cC_{(g)} + dD_{(g)}$$
$$K_p = \frac{(P_C)^c(P_D)^d}{(P_A)^a(P_B)^b}$$

P_A, P_B, P_C, P_D are the partial pressure of gases A, B, C, D and exponen a, b, c, d respectively are the coefficients of balanced equation. Despite the fact that partial pressure of an ideal gas is directly proportional to its molar concentration at constant temperature, the numerical vale of K_c is often not equal to K_p. Constants at a particul constant temperature.

$$K_p = Kc(RT)^{\Delta n}$$

Equilibrium constant some time express in terms of K_n when the nu of moles of reactants and products are given at equilibrium.

$$K_n = \frac{(n_C)^c(n_D)^d}{(n_A)^a(n_B)^b}$$

Here n specify the given number of moles of A, B, C and D in the equilibrium reaction mixture.

5. a) Define battery. Explain the working of lead storage batter determined electrode potential of copper or zinc.

ANSWER:

...TTERY

...battery consists of two or more Galvanic cells connected in series and ...ed to convert chemical energy into electrical energy by means of ...ction-oxidation (redox) reactions.

...d storage battery which is commonly known as car battery is an ...mple of secondary battery.

...12 volt car battery is made up of six Galvanic cells connected in a ...s each of which has an emf of 2 volt. All anode plates are made up ...d (pb) while all cathode plates consists of lead coated with PbO_2. Both ...ps or plates are alternatively suspended into a dilute solution of ...SO_4 of specific gravity 1.25.

...en the battery starts working, lead atoms at anode oxidize to form lead ...s.

$$Pb_{(s)} \longrightarrow Pb^{+2}_{(aq)} + 2e^-(Oxidation\ at\ anode)$$

...e Pb^{+2} ions combine with $SO^{-2}_{4(aq)}$ ions to form solid $PbSO_4$ which is ...osited on anode.

$$Pb^{+2}_{(aq)} + SO^{-2}_{4(aq)} \longrightarrow PbSO_4$$

...electrons generate at anode pushes out into external circuit and then ...circuit into cell from cathode (PbO_2) where PbO_2 reduces to Pb^{+2} ions.

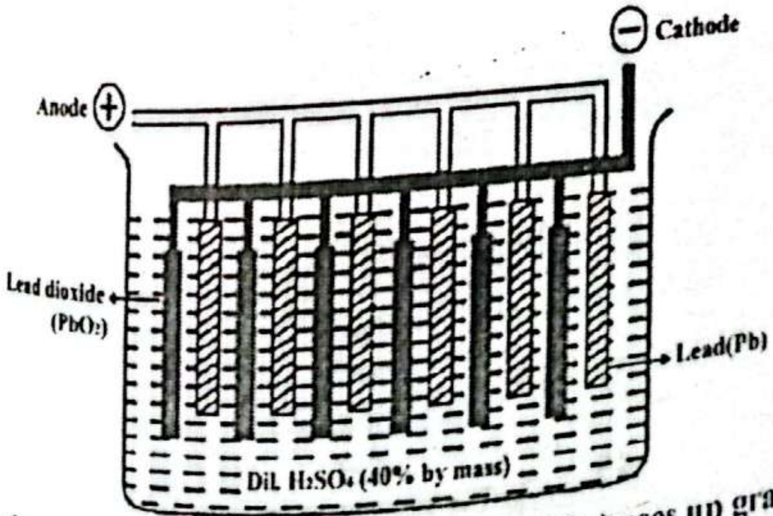
$$PbO_{2(s)} + 4H^+_{(aq)} + 2\bar{e} \longrightarrow Pb^{+2}_{(aq)} + H_2O_{(l)}$$

...e Pb^{+2} ions again combine with SO_4^{-2} ions of electrolyte and produce ...d $PbSO_4$ which is deposited at cathode plates.

...net chemical change at anode and cathode during discharge may be ...ten as.

$$Pb_{(s)} + SO^{-2}_{4(aq)} \rightarrow PbSO_{4(s)} + 2\bar{e}\ (Anode)$$

$$PbO_2 + 4H^+_{(aq)} + SO^{-2}_{4(aq)} + 2\bar{e} \rightarrow PbSO_{4(s)} + H_2O_{(l)}\ (Cathode)$$



...as lead battery works, its electrolyte (H_2SO_4) uses up gradually ...results in converting the electrodes with $PbSO_4$. Thus it becomes unable to release electricity and said to be discharge. This can be ... by lowering in the specific gravity of acid as well as drop of voltage.

b) Give the characteristics of ionic and covalent solids. K_c for re... 55 at 700°C. Calcium the equilibrium concentration of react... products when the concentration of each reactant is 2.55 mole/d...

$$H_2 + I_2 \leftrightharpoons 2HI$$

ANSWER:

DATA:

GIVEN:

Equilibrium constant $= K_C = 55$

Initial concentration of $H_2 = 2.55\ moles/dm^3$

Initial concentration of $I_2 = 2.55\ moles/dm^3$

REQUIRED:

Equilibrium concentration of $H_2 = ?$

Equilibrium concentration of $I_2 = ?$

SOLUTION:

	$H_{2(g)}$	+	$I_{2(g)}$	$\longleftrightarrow$	$2HI$
Moles At initial :	2.55		2.55		0
Moles At equilibrium :	$2.55 - x$		$2.55 - x$		$2x$

$$\because K_C = \frac{[\text{Product}]}{[\text{Reactant}]}$$

$$K_C = \frac{[HI]^2}{[H][I]} = \frac{\left(\frac{No.of\ moles}{Volume}\right)^2}{\left(\frac{No.of\ moles}{Volume}\right)\left(\frac{No.of\ moles}{Volume}\right)}$$

$$K_C = \frac{\left(\frac{2x}{1dm^3}\right)^2}{\left(\frac{2.55-x}{1dm^3}\right)\left(\frac{2.55-x}{1dm^3}\right)}$$

$$49 = \frac{\frac{4x^2}{dm^6}}{\frac{(2.55-x)^2}{dm^6}}$$

$$49 = \frac{4x^2}{(2.55-x)^2}$$

$$49 = \frac{4x^2}{(6.5025 - 5.1x + x^2)}$$

$$49(6.5025 - 5.1x + x^2) = 4x^2$$

$$318.6225 - 249.9x + 49x^2 = 4x^2$$

$$45x^2 - 249.9x + 318.6225 = 4x^2$$

quadratic formula for solving above equation.

$$\because x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a = 45, b = -249.9, c = 318.6225$$

$$x = \frac{-(-249.9) \pm \sqrt{(-249.9)^2 - 4(45)(318.6225)}}{2(45)}$$

$$x = \frac{249.9 \pm \sqrt{62450.01 - 57352.05}}{90}$$

$$x = \frac{249.9 \pm \sqrt{5097.96}}{90}$$

$$x = \frac{249.9 \pm 71.4}{90}$$

Either

$$x = \frac{249.9 + 71.4}{90}$$

$$x = \frac{321.3}{90}$$

$$x = 3.57$$

OR

$$x = \frac{249.9 - 71.4}{90}$$

$$x = \frac{178.5}{90}$$

$$x = 1.98$$

Since the initial moles of H_2 and I_2 are less than 3.57, therefore neglecting this value. so

$$x = 1.9833$$

$Equilibrium\ Concentration\ of\ H_2 = [H_2] = 2.55 - x = 2.55 - 1.98 = 0.57\ moles\ dm^{-3}$

$Equilibrium\ Concentration\ of\ I_2 = [I_2] = 2.55 - x = 2.55 - 1.98 = 0.57 moles\ dm^{-3}$

$Equilibrium\ Concentration\ of\ HI = [HI] = 2x = 2(0.57)$

$$= 114\ moles\ dm^{-3}$$

CHEMISTRY 2023

TIME: 3 Hours **(85 Marks)**

SECTION 'A' (Multiple Choice Questions)(17)

1. Choose the correct answer for each from the given options:

i) Number of atoms in 60gm carbon will be:

* 1.9×10^{23}
* 3.01×10^{24} ✓
* 6.02×10^{23}
* 6.02×10^{24}

ii) Line spectra is used for the identification of a/an:

* Proton
* Electron
* Element
* Molecule✓

iii) According to Hund's rule, the electronic configuration of carbon (C = 6) is:

* $1s^2 2s^2 2p_x^1, 2p_y^1$✓
* $1s^2 2s^2 2p_x^1 2p_y^1 2p_z^1$

* $1s^2 2s^2 p_x^2$ * $1s^2 2s^1 2p_x^1 2p_y^2$

4) The bond order in N_2 molecule is:
* 0 * 1 * 2 * 3✓

5) This molecule has zero dipole moment:
* NH_3 * HCl * H_2O * CCl_4✓

6) Under similar conditions CH_4 gas diffuses faster than SO_2 gas:
* 1.5 times * 2 times✓ * 4 times * 16 times

7) It is an example of crystalline solid:
* Glass * Rubber * Table Salt✓ * Plastic

8) This liquid has the highest surface tension:
* Water * Mercury✓ * Ethanol * Gasoline

9) When a catalyst is added to a system equilibrium K_c will be:
* decreased * increased * unchanged ✓ * zero

10) In equilibrium system $PCl_5 \rightleftharpoons PCl_3 + Cl_2$, the relationship of K_p & K_c is:
* $K_p > K_c$ * $K_p > K_c$ * $K_p = K_c$✓ * $K_p \neq K_c$

11) This oxide is amphoteric in nature:
* K_2O * CO_2 * CaO * Al_2O_3✓

12) In pure water, universal indicator acquires this colour:
* Red * Green ✓ * Blue * Pink

13) Rate constant of a chemical reaction is affected by:
* concentration of reactant * temperature✓
* concentration of product * reaction time

14) The sum of mole fractions of all the components of a solution too:
* 1✓ * 10 * 100 * zero

15) Size of particles in a colloidal solution is:
* less than 1 nm * between 1 nm and 1000 nm
* above 1000 nm * zero

16) In a thermochemical process, work is zero if this is kept constant
* Temperature * Volume✓ * Pressure * Mass

17) The outer body of dry cell is made up of
* Copper * Zinc✓ * Lead * Iron

SECTION 'B' (Short-Answer Questions)

ATOMIC MASS: (H = 1 am, C = 12 amu, Al = 27 amu, O = ... = 35.5 amu, S = 32 amu, Ag = 108amu, N = 14amu)

NOTE: Answer any NINE part questions.
All questions carry equal marks.

2. i) Define any four of the following:
* Molar Volume * Evaporation
* Limiting Reactant * Dipole Moment
* Bond Energy

ANSWER:

MOLAR VOLUME

"The volume of one more of a gas at standard temperature (273K) and pressure (1atm) is referred as molar volume"

EVAPORATION

"At any given temperature, a certain number of the molecules in a liquid possess sufficient kinetic energy to escape from the surface. This process is called evaporation".

BOND ENERGY

"The energy involved for the breaking / formation of one mole of particular type of bonds in the molecule is known as bond energy"

LIMITING REACTANT

"The reactant which is entirely consumed first during chemical reaction is called Limiting reagent"

DIPOLE MOMENT

"The product of magnitude of charge and the distance between positive and negative centers is called dipole moment".

ii) $AlCl_3$ is produced by the following reactant at 500°C

$$2Al + 3Cl_2 \rightarrow 2AlCl_3$$

When $160gm$ of Aluminium is reacted with excess of chlorine gas, $650gm$ of $AlCl_3$ is obtained. Calculate

(a) Theoretical yield (b) % yield

ANSWER:

DATA:

GIVEN:

$Mass\ of\ Aluminum = 160g$

$Atomic\ mass\ of\ Al = 27 \times 1 = 27\ amu$

REQUIRED:

$Theoretical\ yield = ?$

$Percentage\ yield\ of\ AlCl_3 = ?$

SOLUTION:

$$\because Number\ of\ moles = \frac{Given\ Mass}{Molecular\ mass}$$

$$Number\ of\ moles\ of\ Al = \frac{160}{27}$$

$$Number\ of\ moles\ of\ Al = 5.926\ moles$$

MOLE COMPARISON OF Al AND $AlCl_3$

According to Balanced chemical Equation

2 mole of Al gives = 2 mole of $AlCl_3$

1 mole of Al gives = 1 mole of $AlCl_3$

5.926 moles of Al gives = 5.926 moles of $AlCl_3$

$\because$ *Theoretical yield* = *Moles of product* × *Molecular mas*

$$Theoretical\ yield = 5.926 \times 133.5$$

$$Theoretical\ yield = 791.121$$

$$\because Percentage\ yield = \frac{\text{Practical yield}}{Theoretical\ yield} \times 100$$

$$Percentage\ yield = \frac{650}{791.121} \times 100$$

$$Percentage\ yield = 82.161\%$$

iii) Combustion of Ethene (C_2H_4) in air to form CO_2 & H_2O is g the following equation:

$$C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$$

If a mixture containing $2.8\ gm\ C_2H_4$ & $6.4\ gm\ O_2$ is allowed to identify the limiting reactant and determine the mass of CO_2 gas t be formed.

ANSWER:

DATA:

GIVEN:

Given Mass of $C_2H_4 = 2.8g$

Given Mass of $O_2 = 6.4g$

Molecular mass.of $C_2H_4 = 12 \times 2 + 1 \times 4 = 28\ amu$

Molecular mass of $O_2 = 16 \times 2 = 32\ amu$

REQUIRED:

limiting reactant =?

Mass of CO_2 =?

SOLUTION:

$$Number\ of\ moles = \frac{Given\ Mass}{Molecular\ mass}$$

$$Number\ of\ moles\ of\ C_2H_4 = n = \frac{2.8}{28}$$

$$Number\ of\ moles\ of C_2H_4 = n = 0.1\ moles$$

$$Number\ of\ molesof\ O_2 = n = \frac{6.4}{32}$$

$$Number\ of\ molesof\ O_2 = n = 0.2\ moles$$

MOLE COMPARISON OF C_2H_4 AND CO_2

According to Balanced chemical Equation

1 mole of C_2H_4 gives = 2 moles of CO_2

0.1 moles of C_2H_4 gives = 0.2 moles of CO_2

MOLE COMPARISON OF O_2 AND CO_2

According to Balanced chemical Equation

3 mole of O_2 gives = 2 moles of CO_2

1 mole of O_2 gives = $\frac{2}{3}$ mole of CO_2

1 mole of O_2 gives = 0.133 moles of CO_2

Since the number of moles of CO_2 produced by O_2 is less, therefore O_2 is limiting reactant

Amount of NH_3 = $moles\ of\ CO_2 \times Molar\ mass\ of\ CO_2$

Amount of NH_3 = 0.133×44

Amount or mass of NH_3 = $5.852\ g$

(ii) State Pauli's Exclusion Principle. Write electronic configuration of the following:

* $Cl^{-}(Z = 17)$ * $Ca^{+2}\ (Z = 20)$ * $Cu\ (Z = 29)$

ANSWER:

PAULI'S EXCLUSION RULE

Pauli's Exclusion Principle was given by Wolfgang Pouli (1925 A.D) and it is based on experimental observation.

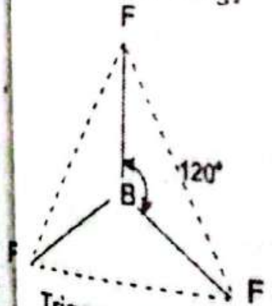
This principle state that, "In an orbital of an atom, no two electrons can have the same set of four quantum numbers, at least one quantum number must be different"

ELEMENTS	CONFIGURATION
$Cl^{-}(Z = 17)$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6$
$Ca^{+2}\ (Z = 20)$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6$
$Cu\ (Z = 29)$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1, 3d^{10}$

(iii) Draw the shape of BF_3, NH_3, CH_4 and H_2O molecules on the basis of VSEPR theory.

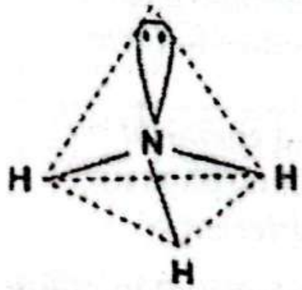
ANSWER:

SHAPE OF BF_3,



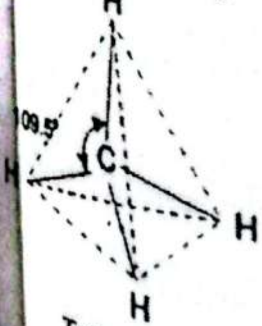
Trigonal Planer

SHAPE OF NH_3



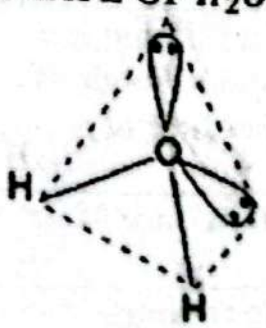
Trigonal Pyramidal

SHAPE OF CH_4



Tetraheral

SHAPE OF H_2O



Angular

vi) Differentiate between any two of the following:
* Crystalline Solid and Amorphous Solid
* Isomorphism and Polymorphism
* Molarity and Molality
* σ bond and π bond

ANSWER:

CRYSTALLINE SOLID AND AMORPHOUS SOLID

CRYSTALLINE SOLID	AMORPHOUS SOLID
They exist in definite geometrical shape	They don't have definite shape
They have sharp melting points.	They melt over a wide range of temperatures
They show anisotropic behavior that is properties like mechanical strength, refractive index and electrical conductivity depends upon direction	They are isotropic in nature
They can be cut down at fixed cleavage plane	When they are cut or hammered cause in irregular fracture

ISOMORPHISM AND POLYMORPHISM

ISOMORPHISM	POLYMORPHISM
The existence of two or more substances of different composition in a similar crystalline form.	The existence or formation of different types of crystal of the same chemical compound.
Isomorphous substance can be crystallized together to form mixed crystals.	Isomorphous substance cannot be crystallized together.
The physical and chemical properties of isomorphous substance are different	The physical properties of different forms of a polymorphous substance are different but chemical properties are same.

MOLARITY AND MOLALITY

MOLARITY	MOLALITY
The molarity of a solution is defined as the total number of moles of solute per liter of a solution.	The molality of a solution is defined as the number of moles of a solute per kilogram of solvent
The units of molarity are M or mol/dm^3.	The units of molality are m or mol/kg.
It depends on the volume.	It doesn't depend on the volume

temperature, and pressure of the solvent.	temperature, and pressure of the solvent.
It increases with an increase in Temperature	It does not change with the changes in the temperature.

SIGMA BOND AND PI BOND

SIGMA BOND	PI BOND
It is formed by head to head overlapping of half filled atomic orbital.	It is formed by lateral/parallel overlapping of half filled atomic orbital.
Only one sigma bond can exist between the bonded atoms.	Maximum two pi bonds can form between the two atoms.
Orbital free rotation can be possible between the two atoms if only sigma bond is present.	Orbital free rotation is not possible between two atoms if pi bond is present.
Electron cloud is denser at the plane of bond axis.	Electron cloud is denser at above and below the plane of bond axis.

(ii) 40 dm^3 of hydrogen gas is collected over water at 23°C and 831 torr pressure. What would be the volume of dry gas obtained at STP (vapour pressure of water at $23°C$ = 21 torr

ANSWER:

DATA:

GIVEN:

$Volume\ of\ moist = V_1 = 40dm^3$

$Initial\ Pressure\ of\ moist = P_1 = 831\ torr$

$Initial\ Temperature = T_1 = 23°C = 23 + 273 = 296K$

$Final\ Temperature = T_2 = 0°C = 0 + 273 = 273K$

$Final\ Pressure = P_2 = 760\ torr$

$Pressure\ of\ water\ vapour = P_W = 21\ torr$

REQUIRED:

$Pressure\ of\ dry\ gas = P_D = ?$

$Volume\ of\ dry\ gas = V_2 = ?$

SOLUTION:

PRESSURE OF DRY GAS

$$P_{Dry\ gas} = P_{Moist\ gas} - P_{Water\ vapours}$$

$$P_D = 831 - 21$$

$$P_D = 810\ torr$$

VOLUME OF DRY GAS AT STP

$$\because \frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$V_2 = \frac{P_1V_1T_2}{T_1P_2}$$

$$V_2 = \frac{(810\,torr)(40dm^3)(273K)}{(296K)(760torr)}$$

$$V_2 = \frac{8845200dm^3}{224960}$$

$$V_2 = 39.319\ dm^3$$

The volume of dry gas at standard condition is $39.319\ dm^3$

viii) Give scientific reasons for the following:

* At Mount Everest, atmospheric pressure falls down to about ... pressure at sea-level.

ANSWER: Since the atmospheric pressure varies with altitude, ... altitude increases, the amount of air over a unit area decreases ... atmospheric pressure will reduce due to lower air molecules. For ... reason at Mount Everest atmospheric pressure falls down to abo... third as compared to the pressure at sea-level

* Water droplets quickly deposit at the outer side of a glass of co...

ANSWER: The glass of cold water tries to cool down the wa... in the air which comes in contact with wall of glass and ... temperature the vapour liquid and appears as water drople... outside of glass, and this phenomenon is known as condensation

* Mercury in a glass tube does not wet the walls.

ANSWER: Mercury does not wet the glass because the cohesi... Mercury is stronger than the adhesive force between the Mercu... glass.

* While boiling an egg, we often mix a small amount of table sal...

ANSWER: NaCl or salt help in reducing the vapour pres... liquid, which in turn increases the boiling point of water, and ... water to attain a higher temperature before starting to boil... added salt to the water prevents an egg from cracking during ...

ix) State Graham's Law of diffusion. The ratio of rates of diffu... gases A and B is 1.5:1 respectively. If the relative molecular ... *A* is 16, find the molecular mass of *B* gas.

ANSWER:

GRAHAM'S LAW OF DIFFUSION

"At constant temperature and pressure, the rate of effusion or ... a gas is inversely proportional to the square root of its density ... mass".

OR

"At a given temperature and pressure the rates of diffusion of ... inversely proportional to the square root of its density".

Mathematically

$$\text{Rate of diffusion} \propto \frac{1}{\sqrt{\text{density}}}$$

$$r \propto \frac{1}{\sqrt{d}}$$

$$r = \frac{k}{\sqrt{d}}$$

TA:

VEN:

lecular mass of $A = 16$

es of diffusion of gases $A = r_1 = 1.5$

es of diffusion of gases $B = r_2 = 1$

EQUIRED:

lecular mass of $B = ?$

UTION:

cording to Graham's Law of Diffusion

$$\because \frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$$

$$\frac{1.5}{1} = \sqrt{\frac{M_2}{16}}$$

uaring on both sides,

$$(1.5)^2 = \left(\sqrt{\frac{M_2}{16}}\right)^2$$

$$2.25 = \frac{M_2}{16}$$

$$2.25 \times 16 = M_2$$

$$M_2 = 36$$

e molecular mass of gas B is 36 amu

Define Solubility Product. Write the solubility product expression for e following:

$PbCl_2$ * $Fe(OH)_3$ * Ag_2SO_4

LUBILITY PRODUCT

e product of ionic concentration when dissolved and undissolved ions e in equilibrium is called the solubility product and its represent by sp.

OR

e product of the concentration of ions in saturated solution of a aringly soluble salt is called solubility product and its represent by Ksp.

SALT	EQUATION OF IONIC EQUILIBRIUM	SOLUBILITY PRODUCT EXPRESSION
$PbCl_2$	$PbCl_{2(s)} \rightleftharpoons Pb^{+2}_{(aq)} + 2Cl^-_{(aq)}$	$K_{sp} = [Pb^{+2}][$
$Fe(OH)_3$	$Fe(OH)_3 \rightleftharpoons Fe^{+3}_{(aq)} + 3OH^-_{(aq)}$	$K_{sp} = [Fe^{+3}][$
Ag_2SO_4	$Ag_2SO_{4(s)} \rightleftharpoons 2Ag^{+2}_{(aq)} + SO^{-2}_{4(aq)}$	$K_{sp} = [Ag^+]^2[$

xi) Define pH and pOH of solution and write its formula. Calcul of $0.0025M$ aqueous solution of HCl at 25°C.

$$HCl_{(l)} \rightleftharpoons H^+_{(aq)} + Cl^-_{(aq)}$$

ANSWER:

pH

"pH is the negative logarithm of the molar concentration of H^+ ion aqueous solution at given temperature".

OR

"The negative logarithm of the concentration of hydrogen ions (H solution is called the pH of a solution".

Mathematically

$$pH = -log[H^+]$$

pOH

"pOH is the negative logarithm of molar concentration of OH^- ion aqueous solution at given temperature".

OR

"The negative logarithm of the concentration of hydroxide ions (O a solution is called the pOH of a solution".

$$pOH = -log[OH^-]$$

DATA:

GIVEN:

Concentration of $HCl = 0.0025M$

REQUIRED:

$pOH = ?$

SOLUTION:

Since HCl is a strong acid, it is assumed to be completely ionized aqueous solution.

$$\underset{(0.0025M)}{HCl_{(l)}} \rightleftharpoons \underset{(0.0025M)}{H^+_{(aq)}} + Cl^-_{(aq)}$$

Therefore for every molecules of HCL, there is one H^+ ion, hence

$$[H^+] = [HCl$$

Or

$$[H^+] = 0.0025M$$

Now using the formula of pH

$\because pH = -\log[H^+]$

$pH = -\log[0.0025]$

$pH = 2.6$

$\because pH + pOH = 14$

$pOH = 14 - pH$

$pOH = 14 - 2.6$

$pOH = 11.4$

Hence the pOH is 11.4

(xii) Calculate standard enthalpy of formation ΔH_f^o of $(C_3H_8O_3)$ from the data given below.

- $3C + 4H_2 + \frac{3}{2}O_2 \rightarrow C_3H_8O_3$ $\quad \Delta H_f^o = ?$
- $C + O_2 \rightarrow CO_2$ $\quad \Delta H_f^o = -393.5KJ/mole$
- $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ $\quad \Delta H_f^o = -285.8KJ/mole$
- $C_3H_8O_3 + 3\frac{1}{2}O_2 \rightarrow 3CO_2 + 4H_2O$ $\quad \Delta H_f^o = -1654.1KJ/mole$

SOLUTION:

$3C_{(s)} + 4H_{2(g)} + \frac{3}{2}O_{2(g)} \longrightarrow C_3H_8O_{3(l)} \quad (\Delta H_f^o = ?)$

$C_{(s)} + O_{2(g)} \longrightarrow CO_{2(g)} \quad (\Delta H^o = -393.5 \text{ KJ/mol}) - - - (i)$

$H_{2(g)} + \frac{1}{2}O_{2(g)} \longrightarrow H_2O_{(l)} \quad (\Delta H^o = -285.8 \text{ KJ/mol}) - - - (ii)$

$C_3H_8O_{3(l)} + 3\frac{1}{2}O_{2(g)} \longrightarrow 3CO_{2(g)} + 4H_2O_{(l)}$

$(\Delta H^o = -1654.1 \text{ KJ/mol}) - - - - (iii)$

Equation (i) multiply by "3" and Equation (ii) multiply by "4", then adding both equations.

$3C_{(s)} + 3O_{2(g)} \longrightarrow 3CO_{2(g)} \quad (\Delta H^o = -1180.5 \text{ KJ/mol})$

$4H_{2(g)} + 2O_{2(g)} \longrightarrow 4H_2O_{(l)} \quad (\Delta H^o = -1143.2 \text{ KJ/mol})$

$3C_{(s)} + 5O_{2(g)} + 4H_{2(g)} \longrightarrow 3CO_{2(g)} + 4H_2O_{(l)}$

$(\Delta H^o = -2323.7 \text{ KJ/mol}) - - - - (iv)$

Subtracting equation (iii) from equation (iv), we get

$3C_{(s)} + 5O_{2(g)} + 4H_{2(g)} \longrightarrow 3CO_{2(g)} + 4H_2O_{(l)} \quad (\Delta H^o = -2323.7)$

$\underset{-}{C_3H_8O_{3(l)}} + \underset{-}{3\frac{1}{2}O_{2(g)}} \longrightarrow \underset{-}{3CO_{2(g)}} + \underset{-}{4H_2O_{(l)}} \quad \underset{+}{(\Delta H^o = -1654.1)}$

$3C_{(s)} + \frac{3}{2}O_{2(g)} + 4H_{2(g)} \longrightarrow C_3H_8O_{3(l)}$

$(\Delta H^o = -669.6 \text{ KJ/mol})$

(xiii) What is order of reaction? Decomposition of NO_2 into NO and O_2 is 2nd order reaction:

$$2NO_2 \rightarrow 2NO + O_2$$

If the rate constant at certain temperature is 3.8×10^{-4} $dm^3 mol$ and initial concentration of NO_2 is 0.38M, calculate the initial r reaction.

ORDER OF REACTION

The order of reaction can be defined as "The sum of all the exponen the concentration in terms of reactants involves in the rate law".

OR

The power of the concentration in the rate law expression is cal order of reaction

DATA:

GIVEN:

Rate constant $= K = 3.8 \times 10^{-4}\, dm^3\, mol^{-1}\, S^{-1}$

$Initial\ concentration\ of\ NO_2 = [NO_2] = 0.38M$

REQUIRED:

$$Initial\ rate\ of\ reaction = \frac{dx}{dt} = ?$$

SOLUTION:

$$\because Rate\ of\ Reaction = \frac{dx}{dt} = K[\text{Reactant}]$$

$$Rate\ of\ Reaction = K[NO_2]^2$$

$$Rate\ of\ Reaction = (3.8 \times 10^{-4}\, dm^3\, mol^{-1}\, S^{-1})(0.38M$$

$$Rate\ of\ Reaction = 5.487 \times 10^{-5} Mol\, dm^{-3}\, S^{-1}$$

xiv) State Raoult's Law. Derive this law mathematically in thre How this law can be applied on binary solution of miscible liquids

ANSWER:

STATE RAOULT'S LAW

A French chemist F.M. Raout studied these phenomena of loweri vapour pressure by the effect of solute particles in a solution and formulated a law (1887) known as Raoult law. For a non-volatile system.

It states as "The vapour pressure of a solution which is made up volatile solute with volatile solvent is directly proportional to the fraction of solvent".

$$P \propto X_1$$

$$P = P^o X_1 \text{ ------- } (i)$$

Here P specifies the vapours pressure of solution, P^o represents pressure of solvent and X_1 is the mole fraction of solvent.

Since in binary solution

$$X_1 + X_2 = 1$$

$$X_1 = 1 - X_2$$

Putting the value of X_1 in eq. (i) we get

$$P = P^o(1 - X_2)$$
$$P = P^o - P^oX_2$$
$$P^oX_2 = P^o - P$$

$$\because \Delta P = P^o - P$$

$$P^oX_2 = \Delta P \text{ - - - - - - - } (ii)$$

Equation (ii) represents the second form of Raoult's law which states as the lowering of vapour pressure is directly proportional to the mole fraction of solute.

$$P^oX_2 = \Delta P$$
$$X_2 = \frac{\Delta P}{P^o} \text{ - - - - - - - } (iii)$$

Here $\frac{\Delta P}{P^o}$ is referred as relative lowering in vapours pressure.

Eq. (iii) represents the third form of Raoult law which state as the relative lowering of vapours pressure is equal to the mole fraction of solute.

Raoult's law is also applicable to a binary solution made up of two miscible volatile liquids having similar order of polarity and molecular size. The partial pressure of each liquid in the mixture is equal to the vapours pressure of the pure component multiplied by its mole fraction and may be written as

$$P_A = P^o_A X_A \; and \; P_B = P^o_B X_B$$

Thus the total pressure of the solution is equal to the sum of the partial pressure of both liquids in the mixture.

$$P_t = P^o_A X_A + P^o_B X_B \text{ - - - - - - } (iv)$$

xiv) OR What is the fourth state of matter? State about its occurrence, characteristics and uses.

FOURTH STATE OF MATTER

The fourth state of matter is called "Plasma". This word is taken from the "Plassein" which means moldable substance. This state can be defined as "The mixture of positive ions, electrons and un-ionized molecules and atoms is known as plasma".

OCCURRENCE OF PLASMA

Natural plasma exists only at high temperature or low temperature vacuums.

CHARACTERISTICS OF PLASMA

1. Natural plasma does not break down or react rapidly but it is extremely hot (over 20000°C).
2. Plasma as a whole is neutral because it contains equal number of negatively charged electrons and positively charged ions.
3. Plasma contains significant number of charged particles which affect its properties.
4. Like gases, plasma has neither definite shape nor volume.

5. Plasma shows characteristics glow in discharge tube depending the nature of the gas. For example hydrogen exhibits green, o... and Nitrogen purple glows.

USES OF PLASMA

Plasma has enormous uses in various fields which are given below
(i) When the light of fluorescent bulb or neon signs is turned on, ... excited and creates glowing plasma which lightens the surround...
(ii) It is used in television and computer chips.
(iii) Plasma is used in rockets propulsion, cleaning the environm... destroying biological hazards and healing wounds.
(iv) It is used in the processing of semiconductors and sterilizati... some medical products.
(v) It is used in lasers, lamp, pulsed power switches and diamo... films.

SECTION "C" (Detailed Answer Questions)

NOTE: Attempt any Two questions from this sec... questions carry equal marks.

3 a) Write the four main postulates of Bohr's atomic theory. ... relation $r_n = \frac{a^o n^2}{z}$ for the radius of n^th^ orbit of Hydrogen atom

ANSWER:

POSTULATE OF BOHR'S THEORY

Postulates of Bohr's theory are given below:

(i) Electrons evolves around the nucleus in circular orbit ... fixed distance from nucleus with definite energy.

(ii) As long as a electrons remains in an appropriate Orbit, ... loses, nor gains energy. Hence each Orbit has a fixed en... however the energy of orbit increases with the increase dis... nuclei.

(iii) During excitations, electrons of absorbs some quan... and jump from a lower energy orbit to an appropriate hi... orbit but when it return back to lower energy orbit it emit... energy.

$$\Delta E = E_2 - E_1 = hv$$

Here "v" is frequency of radiation and "h" is plank const...

(iv) Electron can move only in those orbit in which ... momentum of electrons (mvr) is integral multiple of nh/2π

$$mvr = \frac{nh}{2\pi}$$

Here 'm' and 'v' are the mass of velocity of electron ... radius of orbit.

DERIVATION:

Consider 'e^-' and 'm' are the charge and the mass of the revolving electron around the nucleus, 'Ze' is the charge on the nucleus, and 'r' is the radius of an orbit in which an electron is moving with a velocity 'v'. According to Coulomb's law,

$$F = K\frac{q_1 q_2}{r^2}$$

The above equation for the electrostatic force between the nucleus and electron can be written as:

$$F = K\frac{Ze^+e^-}{r^2}$$

Where K is proportionali constant, it is equal to $\frac{1}{4\pi\varepsilon_o}$

Hence attractive force between the nucleus and electron which is known as the centripetal force can be written as

$$F = \left(\frac{1}{4\pi\varepsilon_o}\right)\frac{Ze^2}{r^2}$$

$$F = \frac{Ze^2}{4\pi\varepsilon_o r^2}$$

But at the same time another force acting on revolving electron which is known as centrifugal force, which is equal to $\frac{mv^2}{r}$

As far as electron remains in the same orbit, the two opposite forces are equal. i.e.

$$Centripetal\,force\,of\,electron = Centrifugal\,force\,of\,electron$$

$$\frac{Ze^2}{4\pi\varepsilon_o r^{\cancel{2}}} = \frac{mv^2}{\cancel{r}}$$

$$mv^2 r = \frac{Ze^2}{4\pi\varepsilon_o}$$

$$r = \frac{Ze^2}{4\pi\varepsilon_o mv^2} - - - (i)$$

According to Bohr's postulate;

Angular momentum of electron is must be an integral multiple of= $\frac{nh}{2\pi}$ i.e.

$$\therefore\ mvr = \frac{nh}{2\pi}$$

$$v = \frac{nh}{2\pi\, mr}$$

Putting the value of v in equation (i)

$$r = \frac{Ze^2}{4\pi\varepsilon_o m\left(\frac{nh}{2\pi\, mr}\right)^2}$$

$$r = \frac{Ze^2}{4\pi\varepsilon_o m\left(\frac{n^2h^2}{4\pi^2m^2r^2}\right)}$$

$$r = \frac{Ze^2}{\varepsilon_o\left(\frac{n^2h^2}{\pi mr^2}\right)}$$

$$r = \frac{\pi Ze^2mr^2}{\varepsilon_o n^2h^2}$$

$$\frac{r^2}{r} = \frac{\varepsilon_o n^2h^2}{Z\pi me^2}$$

$$r = \frac{n^2\varepsilon_o h^2}{Z\pi me^2} --- (ii)$$

Where

ε_o *Vacuum permittivity constant* $= 8.84 \times 10^{-12} C^2 J^{-1} m$

$h =$ *plank constant* $= 6.625 \times 10^{-34} J.S$

$m =$ *mass of electron* $= 9.11 \times 10^{-31} Kg$

$e =$ *charge of electron* $= 1.602 \times 10^{-19} C$

Assembling all constants in equation(*iii*), we get

$$r = \frac{n^2(8.84 \times 10^{-12})(6.625 \times 10^{-34})^2}{Z\left(\frac{22}{7}\right)(9.11 \times 10^{-31})(1.602 \times 10^{-19})^2}$$

$$r = \frac{n^2(8.84 \times 10^{-12})(43.89 \times 10^{-68})}{Z(3.142)(9.11 \times 10^{-31})(2.566 \times 10^{-38})}$$

$$r = \frac{n^2(8.84 \times 43.89 \times 10^{-12-68})}{Z(3.142 \times 9.11 \times 2.566 \times 10^{-31-38})}$$

$$r = \frac{n^2(387.99 \times 10^{-80})}{Z(73.45 \times 10^{-69})}$$

$$r = \frac{n^2}{Z}(5.282 \times 10^{-11})$$

$$r = \frac{n^2}{Z}(0.5282 \times 10^{-10} m)$$

$$r = \frac{n^2}{Z}\left(\frac{0.529 \times 10^{-11}}{10^{-10}} m\right)$$

$$r = \frac{n^2}{Z}(0.529 \mathring{A})$$

$$r = \frac{n^2}{Z}(a^o)$$

$$r = \frac{a^o n^2}{Z}$$

$$r_n = \frac{a^o n^2}{Z}$$

It is the required equation used to determination of the radius of the nth orbit of a hydrogen atom. Where n is the number of orbits, Z is the atomic number and a^o is Bohr's constant and its value is 529A

Q (b) What is Hybridization? Write its types. CH_4, NH_3 & H_2O all have same hybridization but their molecular structures are different. Explaining their structures give a reason for this difference.

HYBRIDIZATION

The mixing of different atomic orbital to produce the same number of equivalent orbital, having same shape and energy is known as hybridization".

TYPES OF HYBRIDIZATION

sp^3 -HYBRIDIZATION:

"Combination of one s and three p orbitals to produce four sp^3 hybrid orbitals is known as sp^3 or tetrahedral hybridization".

sp^2 -HYBRIDIZATION:

"The mixing of one s and two p orbital to produce three sp^2 hybrid orbital is referred to as sp^2 (trignoal hybridization)".

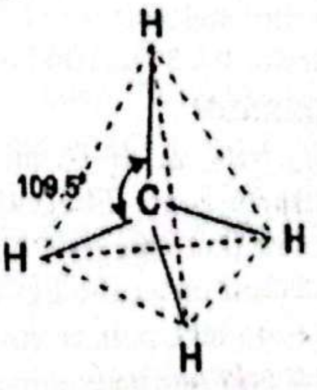
sp - HYBRIDIZATION:

The hybridization of one s and one p atomic orbital leads to two hybrid orbital known as sp hybrid orbital

STRUCTURES OF METHANE MOLECULES (CH_4)

The electronic configuration of carbon is ($1s^2$, $2s^2$, $2px^1$, $2py^1$, $2pz^0$) but it is assumed that one electron of 2s orbital get promoted to 2pz orbital to make it tetravalent. C (z = 6) $1s^2$, $2s^2$, $2px^1$, $2py^1$, $2px^1$ (tetravalent)

In Methane, the four atomic orbital of carbon undergo a mixing process to produce four new orbital of equal energy and shapes, each named as sp^3 hybrid orbital. All four sp^3 hybrid orbital then directed towards the corner of a regular tetrahedon at angle of 109.5°. Since each sp^3 hybrid orbital has one electron, it overlap with s orbital of hydrogen atom. Thus four $C-H$ sigma bonds are formed.

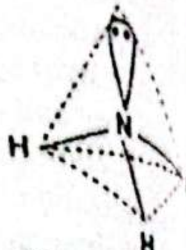


Tetraheral

STRUCTURES OF AMMONIA MOLECULE (NH_3)

The electronic configuration of Nitrogen is $1s^2$, $2s^2$, $2px^1$, $2py^1$, $2pz^1$. Thus according to the concept of hybridization one s and three p orbital of the valence shell of nitrogen undergo the processes of mixing to give four sp^3 hybrid orbital which are directed towards the four corners tetrahedron.

Out of four hybrids orbital three have single electrons hence they overlap with s-orbital of hydrogen and form three $(N-H)$ sigma bond. The fourth sp^3 orbital which has lone pair of electron remains unbounded. The stronger repulsion of this orbital deviate the bond angle form 109.5^{o} to 107^{o} and give rise to a pyramidal geometry.

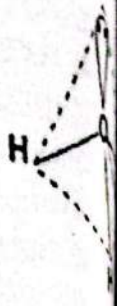


Trigonal Pyra

STRUCTURES OF WATER MOLECULES (H_2O)

The electronic configuration of gas is $1s^2$, $2s^2$, $2px^2$, $2py^1$, $2pz^1$. The four outer shells orbitals are assumed to be mixed together which results in the formation of four new sp^3 hybrid orbitals which are directed at the corner of tetrahedron.

Since only two sp^3 orbitals possess signal electrons, they overlap with s-orbital of hydrogen to from two $O-H$ sigma bonds. The remaining two sp^3 hybrid orbital contain lone pairs of electrons.



Angu

Since only two sp^3 orbitals possess signal electrons, they overlap w orbital of hydrogen to from two $O-H$ sigma bonds. The remaining sp^3 hybrid orbital contain lone pairs of electrons.

Because the repulsion of lone pair is greater than bond pair, the sh water molecule is not regular tetrahedron. The distortion of bond a from 109.5^{o} to 104^{o} makes an angular geometry.

REASON

H_4, NH_3 & H_2O all have same hybridization but their m structures are different because of difference of lone and bond pa know that lone pair has greater repulsion as compare to bond p all these molecules has different numbers of lone and bond pairs has no lone pair so its shape is regular tetrahedron, similarly has only one lone pair and three bond pair so it has pyramidal while water has two lone pair and two bond pair so it makes a geometry rather than regular tetrahedron.

4 a) State Hess's Law? What is meant by Born-Haber's Cycle? the lattice energy for $CsF_{(s)}$ writing all the involved therm equations stepwise, using following data: (all values are in KJ/mole)

* ΔH_f^o of CsF = -553.5

* $\Delta H_{(sub)}$ of Cs = 76

* $\Delta H_{(sub)}$ of Cs = 375.5

ΔH_{EA} of F = -328.2

ΔH_{Diss} of F_2 = 157

ANSWER:

HESS'S LAW

G.H Hess (1840) introduced a law of thermochemistry based on his observations on cyclic thermochemical changes which involves two or more different routes. This is known as Hess's law of enthalpy summation.

STATEMENT

This law states as "If a chemical reaction can be brought about in more than one path way, the net enthalpy change is the same provided that the initial and final states are the same".

BORN HABER CYCLE

Born Haber cycle is developed by two German chemists Max Born and Fritz Haber (1919) to correlate the lattice energy of ionic solids with some other energies involved in their formation.

"It is a cycle of enthalpy change in which and ionic solid is theoretically formed from its elements in their standard states".

It is a specific application of Hess's law and generally use to determine lattice energy of ionic solids.

Determination of the lattice energy of $CsF_{(s)}$

STEP-1: SUBLIMATION OF CESIUM

It involves the vapourization of metallic cesium into gaseous atom. The process is endothermic because heat is always needed to sublime a solid.

$$Cs_{(s)} \rightarrow Cs_{(g)} \quad (\Delta H_{(sub)} = 76.5\ KJ/mol)$$

STEP-2: IONIZATION OF CESIUM ATOM

It involves the loss of electron from cesium gaseous atom. The enthalpy change corresponds to this process is always positive and known as ionization enthalpy.

$$Cs_{(g)} \rightarrow Cs^{+}_{(g)} + e^{-} \quad (\Delta H_{(IE)} = 375.5KJ/mol)$$

STEP-3: DISSOCIATION OF GASEOUS FLUORINE

It involves the breaking of F_2 molecules into fluorine gas atoms. The enthalpy related to this process is ever positive and labeled as enthalpy of dissociation or often known as enthalpy of atomization.

$$\frac{1}{2} F_{2(g)} \rightarrow F_{(g)} \quad (\Delta H_{(D)} = \frac{1}{2} \times 157\ KJ/mol)$$

STEP-4: FORMATION OF F^{-} ION

It involves the absorption of electron into fluorine gas atom. The enthalpy change of this thermochemical process is negative and specify as electron affinity.

$$F_{(g)} + e^- \rightarrow F^-_{(g)} \qquad (\Delta H_{(EA)} = -328.2\ KJ/m$$

STEP=5: FORMATION OF SOLID CsF

It involves the electrostatic interaction of oppositely charged gas The enthalpy change in this process is indeed negative and know Lattice energy.

$$Cs^+_{(g)} + F^-_{(g)} \rightarrow CsF_{(s)} \qquad (\Delta H_{(LE)} = ?)$$

Intended with Hess's Law, the sum of enthalpy change of these fi will be equal to the enthalpy of formation of $CsF_{(s)}$ bring the overa enthalpy change of the energy cycle is zero.

$$\Delta H^o_f = \Delta H_{sub} + \Delta H_{(LE)} + \frac{1}{2}\Delta H_{(D)} + \Delta H_{(EA)} + \Delta H_{(IE}$$

Substituting the values of concerned energies, we get lattice energ

$$(-553.5) = (76.5) + (375.5) + \left(\frac{1}{2} \times 157\right) + (-328.2) + (-$$

$$\Delta H_{LE} = 755.8KJ/mol$$

4 b) Give the statement and mathematical derivation of First L Thermodynamics and derive $q_v = \Delta E$ and $q_p = \Delta H$

ANSWER:

FIRS LAW OF THERMODYNAMICS

"In any physical or chemical process, the net energy of a system its surroundings must remain constant".

OR

"A system can not create or destroy energy, however, it can exc energy with its surrounding in the form of heat and work".

PROOF:

Consider a gas cylinder having a frictionless piston containing gas. The cross-section area of the piston is "A" and "P" is the pre (Force per unit area=F/A) applied by gas on the piston. If q amou heat is supplied to the system, its piston covers some distance Δh i upward direction and the System perform some work i.e.

$$q = \Delta E + W$$

If the piston of the cylinder is fixed, then the change in internal en is equal to the amount of heat supplied to the system. At cons volume, the system cannot be able to do some work.

Mathematically

AT CONSTANT VOLUME (q_v)

$$W = 0$$

Applying first law of thermodynamic

$$\Delta E = q - W$$

$\because$ *At constant Volume* $q = q$

$$\Delta E = q_v - 0$$
$$\Delta E = q_v$$

is represent the amount of heat absorbed by the system at constant lume.

ROOF:

know that

$$q = \Delta E + P\Delta V$$

$$\because \Delta E = E_2 - E_1 \quad and \quad \because \Delta V = V_2 - V_1$$

ut At constant Pressure $q = q_P$

$$\therefore q_P = E_2 - E_1 + P(V_2 - V_1)$$
$$q_P = E_2 - E_1 + PV_2 - PV_1$$
$$q_P = E_2 + PV_2 - (E_1 + PV_1)$$
$$\because H = E + PV$$
$$\therefore q_P = H_2 - H_1$$
$$q_P = \Delta H$$

) Define Oxidation and Reduction according to modem electronic cept. Balance the following equation by ion electron method:

$Cr_2O_7^{-2} + Cl^- \rightarrow + Cr^{+3} + Cl_2$ (Acidic medium)

$MnO_4^- + SO_3^{-2} \rightarrow Mn^{+2} + SO_4^{-2}$ (Basic medium)

SWER:

IDATION

cording to the modern electronic concept, "Oxidation is the mical process in which electrons are lost by an atom or ion".

OR

cording to the oxidation number concept "Oxidation is a process in ich the oxidation state of an element increases".

DUCTION

cording to the modern electronic concept, "reduction is the chemical cess in which atom or ion gain electron".

OR

cording to the oxidation number concept "Reduction is a process in ich the oxidation state of an element decreases".

$Cr_2O_7^{-2} + Cl^- \longrightarrow Cr^{+3} + Cl_2$ (Acidic medium)

LUTION:

$Cr_2O_7^{-2} + Cl^- \longrightarrow Cr^{+3} + Cl_2$

$Cr_2O_7^{-2} \longrightarrow Cr^{+3}$

$Cl^- \longrightarrow Cl_2$

Oxidation +6 to +3

Reduction -1 to +0

$Cr_2O_7^{-2} + 14H^+ \longrightarrow 2Cr^{+3} + 7H_2O$

$Cl^- \longrightarrow Cl_2$

$Cr_2O_7^{-2} + 14H^+ + 6e \longrightarrow 2Cr^{+3} + 7H_2O$

$2Cl^- \longrightarrow Cl_2 + 2e$

$[Cr_2O_7^{-2} + 14H^+ + 6e \longrightarrow 2Cr^{+3} + 7H_2O] \times 1$

$[2Cl^- \longrightarrow Cl_2 + 2e] \times 3$

$Cr_2O_7^{-2} + 14H^+ + 6e \longrightarrow 2Cr^{+3} + 7H_2O$

$6Cl^- \longrightarrow 3Cl_2 + 6e$

Adding both equations

$Cr_2O_7^{-2} + 14H^+ + \cancel{6e} \longrightarrow 2Cr^{+3} + 7H_2O$

$6Cl^- \longrightarrow 3Cl_2 + \cancel{6e}$

$$Cr_2O_7^{-2} + 6Cl^- + 14H^+ \longrightarrow 2Cr^{+3} + 3Cl_2 + 7H_2O$$

$$MnO_4^- + SO_3^{-2} \longrightarrow Mn^{+2} + SO_4^{-2} \text{ (basic medium)}$$

SOLUTION:

$MnO_4^- + SO_3^{-2} \longrightarrow Mn^{+2} + SO_4^{-2}$

$SO_3^{-2} \longrightarrow SO_4^{-2}$ Oxidation +4 to +6

$MnO_4^- \longrightarrow Mn^{+2}$ Reduction +7 to +2

$SO_3^{-2} + 2OH^- \longrightarrow SO_4^{-2} + H_2O$

$MnO_4^- + 4H_2O \longrightarrow Mn^{+2} + 8OH^-$

$SO_3^{-2} + 2OH^- \longrightarrow SO_4^{-2} + H_2O + 2e$

$MnO_4^- + 4H_2O + 5e \longrightarrow Mn^{+2} + 8OH^-$

$[SO_3^{-2} + 2OH^- \longrightarrow SO_4^{-2} + H_2O + 2e] \times 5$

$[MnO_4^- + 4H_2O + 5e \longrightarrow Mn^{+2} + 8OH^-] \times 2$

$5SO_3^{-2} + 10OH^- \longrightarrow 5SO_4^{-2} + 5H_2O + 10e$

$2MnO_4^- + 8H_2O + 10e \longrightarrow 2Mn^{+2} + 16OH^-$

Adding both equations

$5SO_3^{-2} + 10OH^- \longrightarrow 5SO_4^{-2} + 5H_2O + \cancel{10e}$

$2MnO_4^- + 8H_2O + \cancel{10e} \longrightarrow 2Mn^{+2} + 16OH^-$

$$5SO_3^{-2} + 2MnO_4^- + 3H_2O \longrightarrow 5SO_4^{-2} + 2Mn^{+2} + 6$$

5 b) Define Cell-Potential, Standard Electrode Potential. Exp determination of standard electrode Zinc with diagram and po cell reactions.

ANSWER:

CELL POTENTIAL

The electrical energy produce by voltaic cell is due to the elec from anode to cathode through an external wire. "The driv with which the electrons pushes out from anode into external known as electromotive force (emf) or cell potential or cell vol

...ANDARD ELECTRODE POTENTIAL

...standard electrode potential of a single half-cell is defined as ...difference of potential created between the metal electrode and 1 ...r solution of its ions at 25°C and 1 atmospheric pressure".
...abbreviated by E^o and measured in volts.

...TERMINATION OF STANDARD ELECTRODE ZINC

...ndard electrode potential of an electrode can be measured by using ...ference electrode such as SHE. Let consider we want to determine ...electrode potential of zinc. For this purpose we have to construct a ...nic cell made of zinc-SHE electrodes. The first half cell consists of a ...p of zinc immersed in 1M $ZnSO_4$ solution while the other half cell ...sists of platinum foil immersed in 1 M H_2SO_4 solution at standard ...ditions.

...alt bridge is made of KCl jelly which completes the circuit between ...two half cells and prevent the mixing of solutions. Both electrodes ...connected to each other through an external wire by means of a ...ltmeter.

...enever the construction is completed and Galvanic cell is ...erational, the voltmeter shows emf of the cell which comes out to be ...6 volt. Now note the flow of electrons in the external circuit. It ...uld be from zinc electrode to SHE. Thus electrons must have ...ginated at zinc and zinc is anode where oxidation takes place.
...lf cell reactions on zinc and SHE electrode are written as

$$Zn_{(s)} \rightarrow Zn^{+2}_{(aq)} + 2\bar{e} \quad (\textit{Oxidation at anode})$$

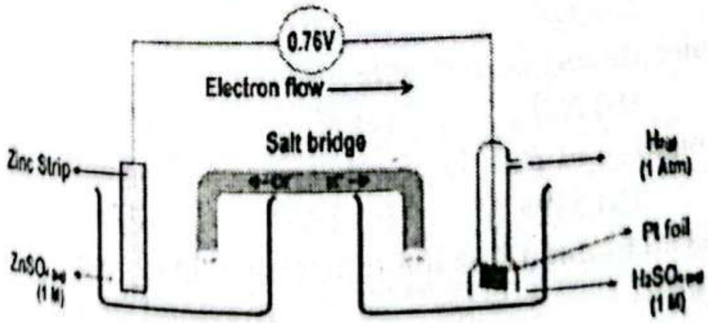
$$H^+ + 2\bar{e} \rightarrow H_2 \quad (\textit{Reduction at cathode})$$

...e electrode potential at zinc is now determined as

$$E^o_{cell} = E^o_{(cathode)} - E^o_{(anode)}$$

$$0.76 = (0.0) = E^o_{(Zn)}$$

$$E^o_{(Zn)} = -0.76\ volt$$



NEW MODEL QUESTION PAPER

Annual Examination 2023

CHEMISTRY (THEORY)

CLASS XI (SCIENCE)

Total Time: 3 Hours **Total Marks: 85**

SECTION 'A' (COMPULSORY)

MULTIPLE CHOICE QUESTIONS (M.C.Qs)

NOTE: This Section consists of 33 MCQs and all are to be answered. Each MCQ carry 1 marks.

1. Choose the correct answer from the given options:

(i) The total number of ions in one formula unit of $FeCl_3$ are:
(a) 6.02×10^{23} (b) 12.04×10^{23}
(c) 18.06×10^{23} (d) 24.08×10^{23} ✓

(ii) When 7.0×10^{12} is multiplied by 2.0×10^{-3}, the answer will be:
(a) 1.4×10^{9} (b) 1.4×10^{10} ✓
(c) 1.4×10^{-15} (d) 1.4×10^{-36}

(iii) Quantum number value for 3d orbitals are:
(a) $n = 2, \ell = 1$ (b) $n = 3, \ell = 2$ ✓
(c) $n = 3, \ell = 3$ (d) $n = 2, \ell = 3$

(iv) The range of wavelength of x-rays lies between;
(a) $0.1Å$ *to* $10Å$ ✓ (b) $10Å$ to $100Å$
(c) $100Å$ *to* $500Å$ (d) $4000Å$ *to* $7000Å$

(v) Bohr's theory cannot be applied on:
(a) H (b) H^{+}✓ (c) He^{+1} (d) Li^{+2}

(vi) What is the hybrid state of carbon in C_2H_2 molecule:
(a) sp^3 (b) sp^2 (c) sp✓ (d) dsp^2

(vii) This molecule has zero dipole moment:
(a) C_6H_6✓ (b) NH_3 (c) H_2S (d) NO_2

(viii) The geometry of BF_3 is planar trigonal, its bond angle should be:
(a) $104.5°$ (b) $109.5°$ (c) $107°$ (d) $120°$ ✓

(ix) VBT tells us about all of the following facts except:
(a) Bond length (b) Bond strength
(c) Bond energy (d) Bond order

(x) Cooling appliances like air conditioners and refrigerators are working on the principle of:
(a) Common ion effect (b) Joule-Thomson effect ✓

(c) Pauli's exclusion principle (d) Le-Chatlier's principle

The rate of diffusion of Helium (He) compared with CH_4 is:

(a) Same time (b) Two times✓

(c) Three times (d) Four times

The molar volume of Oxygen (O_2) is highest at:

(a) 0°C and 1 atm (b) $25°C$ and 2 atm

(c) 40°C and 0.5 atm ✓ (d) $40°C$ and 0.5 atm

Plasma is the fourth state of matter, it consists of:

(a) Neutral molecules (b) Positive ions

(c) Negative electrons (d) All of these ✓

Cooking time is reduced in a pressure cooker because:

(a) Boiling point of water rises✓

(b) Heat is stored in pressure cooker

(c) Vapor pressure of liquid is reduced

(d) Heat is uniformly distributed

Which of the following molecule possess strongest London forces:

(a) H_2 (b) He (c) CH_4 ✓ (d) Ne

Which of the following pair of compounds may represents isomorphism:

(a) $NaCl$ and KNO_3 (b) MgO and NaF ✓

(c) $NaNO_3$ and CdS (d) NaF and $CaCO_3$

(ii) A big crystal can be cut or split into smaller size of identical type; this phenomenon is called:

(a) Anisotropy (b) Cleavage ✓

(c) Symmetry (d) Isomorphism

(iii) $Kp = Kc$ when Δn is equal to:

(a) zero ✓ (b) 1 (c) -1 (d) 2

(x) The solubility of $MgCl_2$ is X, its Ksp will be:

(a) x^2 (b) $2x^2$ (c) $4x^2$ (d) $4x^3$ ✓

(i) The unit of rate constant for the first order reaction is:

(a) Ms^{-1} (b) s^{-1} ✓ (c) $M^{-1}\,s^{-1}$ (d) $M^{-2}\,s^{-1}$

(ii) Amphoteric substance among the following is:

(a) K_2O (b) CO_2 (c) ZnO✓ (d) MgO

(iii) Which of the following salt is hydrolyzed in water:

(a) Na_2SO_4 (b) KCl (c) NH_4Cl ✓ (d) $NaNO_3$

(iiii) Conjugate base of HCO_3^- is:

(a) H_2CO_3 (b) CO_3^{-2} ✓ (c) H^+ (d) H_2O

(xiv) The decomposition of H_2O_2 is inhibited by:

(a) Ethanol (b) Glycerine✓ (c) MnO_2 (d) V_2O_5

(xv) The rate constant of a reaction depends upon:

(a) Temperature ✓ (b) Initial concentration
(c) Time of reaction (d) Extent of reaction

(xxvi) Effect of pressure change play significant role in the solubi of:
(a) Solid into liquid (b) Liquid into liquid
(c) Gas into liquid✓ (d) All of them

(xxvii) Milk is an example of this type of colloid:
(a) Gel (b) Aerosol (c) Emulsion✓ (d) Foam

(xxviii) Parts per trillion means:
(a) 10^3 (b) 10^6 (c) 10^9 (d) 10^{12} ✓

(xxix) Which of the following enthalpy change is always negative:
(a) Enthalpy of formation (b) Enthalpy of decomposi
(c) Enthalpy of combustion✓ (d) Enthalpy of reaction

(xxx) Which of the following is not a state function of a system?
(a) Pressure (b) Enthalpy ✓
(c) Internal energy (d) Work done

(xxxi) Oxidation number of Cr in $Na_2Cr_2O_7$ is:
(a) + 3 (b) + 6 ✓ (c) + 8 (d) + 12

(xxxii) Galvanized rode of iron is coated with:
(a) Nickel (b) Zinc✓ (c) Chromium (d) Carbon

(xxxiii) KOH is used as electrolyte in:
(a) Lead accumulator (b) Fuel cell
(c) Alkaline battery✓ (d) Dry cell

SECTION 'B' (SHORT-ANSWER QUESTIONS

(Marks: 4x8=32)

NOTE: Answer any Eight (8) questions from this section. All questions carry equal marks.

Q.2 (i) (a) What is meant by actual yield? Why it is always less theoretical yield in a reaction.

ANSWER:

ACTUAL YIELD

The actual amount of product which is formed in experiment is practical or actual yield.

Usually, the actual yield is lower than the theoretical yield becau reactions truly proceed to completion (i.e., aren't 100% effici because not all of the product in a reaction is recovered.

Q.2 (b) The volume of a sample of Nitrogen gas (N_2) at STP is 11.2 calculate the mass and number of molecules of N_2 in the sample.

ANSWER:

DATA:

GIVEN:

Volume of $N_2 = 1120cm^3 = \frac{1120}{1000} = 1.12dm^3$

Molecular mass of $N_2 = 28$

REQUIRED:

Mass of $N_2 = ?$

No. of moles of $N_2 = ?$

SOLUTION:

We know that

$22.4\ dm^3 of\ N_2 = 1\ mole$

$1\ dm^3 of\ N_2 = \frac{1}{22.4}\ mole$

$1.12\ dm^3 of\ N_2 = \frac{1}{22.4} \times 1.12\ mole$

$1.12\ dm^3 of\ N_2 = \frac{1}{22.4} \times 1.12\ mole$

$1.12\ dm^3 of\ N_2 = 0.05\ moles$

We know that

$22.4\ dm^3 of\ N_2 = 28\ grams$

$1\ dm^3 of\ N_2 = \frac{28}{22.4}\ grams$

$1.12\ dm^3 of\ N_2 = \frac{28}{22.4} \times 1.12\ grams$

$1.12\ dm^3 of\ N_2 = 1.4\ grams$

2. (ii) Aluminum Sulphide is prepared by the reaction of Aluminum metal and sulphur powder at elevated temperature.

$$2Al + 3S \rightarrow Al_2S_3$$

If 135g Aluminum and 160g sulphur are taken for the reaction, calculate what mass of Al_2S_3 will be formed.

ANSWER:

DATA:

GIVEN:

Given mass of Al = 135g

Given mass of S = 160g

Atomic mass of $Al = 27amu$

Atomic mass of $S = 32amu$

Molecular mass of $Al_2S_3 = 27 \times 2 + 32 \times 3 = 54 + 96 = 150amu$

REQUIRED:

Mass of $Al_2S_3 = ?$

SOLUTION:

$$Number\ of\ moles = \frac{Given\ Mass}{Molecular\ mass}$$

$$Number\ of\ moles\ of\ Al = n = \frac{135}{27}$$

$$Number\ of\ moles\ of\ Al = n = 5\ moles$$

$$Number\ of\ molesof\ S = n = \frac{160}{32}$$

$$Number\ of\ molesof\ S = n = 5\ moles$$

MOLE COMPARISON OF Al AND Al_2S_3

According to Balanced chemical Equation

2 mole of Al gives = 1 mole of Al_2S_3

1 mole of Al gives = $\frac{1}{2}$ mole of Al_2S_3

5 moles of Al gives = $\frac{5}{2}$ moles of Al_2S_3

5 moles of Al gives = 2.5 moles of Al_2S_3

MOLE COMPARISON OF S AND Al_2S_3

According to Balanced chemical Equation

3 mole of S gives = 1 mole of Al_2S_3

1 mole of S gives = $\frac{1}{3}$ mole of Al_2S_3

5 moles of S gives = $\frac{5}{3}$ moles of Al_2S_3

5 moles of S gives = 1.66 moles of Al_2S_3

Since the number of moles of Al_2S_3 produced by S is less, therefore S is limiting reactant

Amount of Al_2S_3 = $moles\ of\ Al_2S_3 \times Molar\ mass\ of\ Al_2S_3$

Amount of $Al_2S_3 = 1.66 \times 150$

Amount or mass of $Al_2S_3 = 249g$

2. (iii) State Pauli and Hund's rule. Write the electronic configuration of the following species:

* Ca^{+2} $(Z = 20)$ * Br^{-1} $(Z = 35)$

ANSWER:

PAULI'S EXCLUSION RULE

Pauli's Exclusion Principle was given by Wolfgang Pouli (1925 A.D) it is based on experimental observation.

This principle state that, "In an orbital of an atom, no two electrons can have the same set of four quantum numbers, at least one quantum number must be different"

HUND'S RULE OF MAXIMUM MULTIPLICITY

This principle state that, "In available degenerated orbitals (p, d and f) electrons are distributed in such a way that the maximum number of half filled orbitals (single electron in orbitals) are obtained".

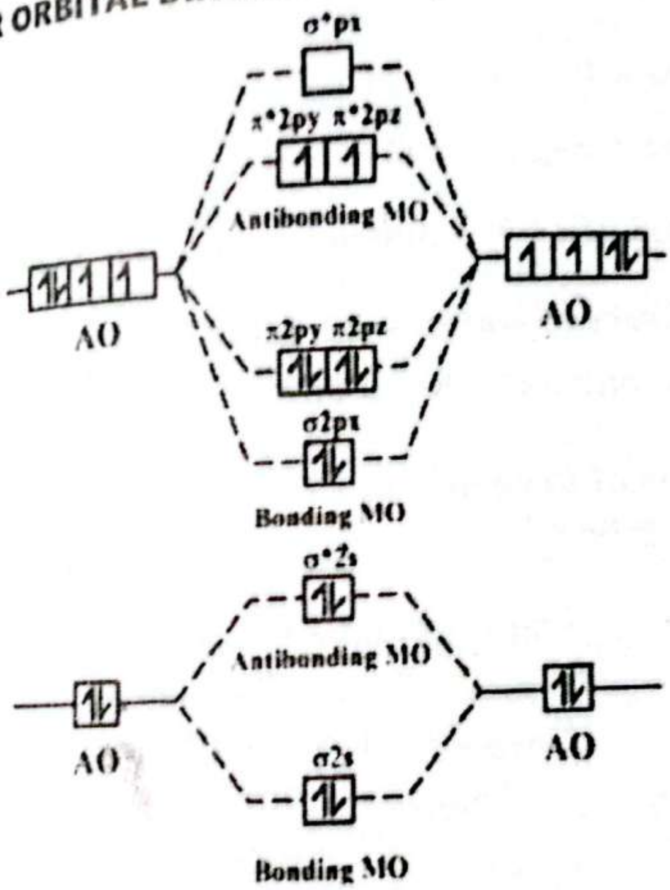
ECTRONIC CONFIGURATION

ELEMENTS	CONFIGURATION
Ca^{+2} $(Z = 20)$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6$
Br^{-1} $(Z = 35)$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}, 4p^6$

) Draw molecular orbital diagram of O_2 molecule. Find bond order O_2 molecule and explain why O_2 molecule is paramagnetic? (9)

SWER:

ECULAR ORBITAL DIAGRAM OF O_2



D ORDER

order of O_2 molecule in determine as

ond order

$$= \frac{\text{Electrons in bonding M.O} - \text{Electrons in anti bonding M.O}}{2}$$

$$= \frac{N(b) - N(a)}{2}$$

$$\text{Bond order of } O_2 = \frac{10 - 6}{2} = \frac{4}{2} = 2$$

the bond order of O_2 molecule is 2.

AMAGNETIC BEHAVIOR OF OXYGEN MOLECULE

ompounds or molecules which have at least an unpaired electron in orbitals are called paramagnetic substances. In an oxygen molecule.

there are two unpaired electrons in its molecular orbital, which i shows paramagnetic behaviour.

2. (v) Oxygen gas was collected over water at 24°C and a total pres 762 *torr*. If the volume of the gas collected was $300cm^3$. Calculate number of moles and the mole fraction of oxygen gas in the mixture vapour pressure of water at 22.4 *torr*).

ANSWER:

DATA:

GIVEN:

$$Temperature = T = 24°C = 24 + 273 = 297K$$

$$Total\ Pressue = P_{Total} = 762\ torr = \frac{762}{760} = 1atm$$

$$Volume\ of\ Oxygen = V = 300cm^3 = \frac{300}{1000} = 0.3dm^3$$

$$Vapour\ pressure\ of\ water = 22.4\ torr = \frac{22.4}{760} = 0.0925atm$$

$$General\ Gass\ constant = R = 0.0821\ atm.dm^3/mol.K$$

REQUIRED:

$$Mole\ fraction\ of\ oxygen = X_o = ?$$

$$No.of\ moles = n_O = ?$$

SOLUTION:

$$\because P_{Total} = P_{Oxygen} + P_{Water\ Vapours}$$

$$P_{Oxygen} = P_{Total} - P_{Water\ Vapours}$$

$$P_{Oxygen} = 1atm - 0.0925atm$$

$$P_{Oxygen} = 0.9075\ atm$$

$$\because PV = nRT$$

$$PV = \left(\frac{m}{M}\right)RT$$

$$m = \frac{MPV}{RT}$$

$$m = \frac{(32)(0.9075)(0.3)}{(0.0821)(297)} = \frac{8.712}{24.3837}$$

$$m = 0.3572g$$

$$\because No.of\ Moles = \frac{Given\ Mass}{Molecular\ Mass}$$

$$No.of\ Moles\ of\ O_2 = \frac{0.3572}{32}$$

$$No.of\ Moles\ of\ O_2 = 0.0116\ moles$$

$$\because Mole\ Fraction = \frac{Pressure\ of\ gas}{Total\ pressure\ of\ Mixture}$$

$$X_{Oxygen} = \frac{Pressure\ of\ Oxygen}{Total\ pressure\ Mixture} = \frac{0.9075}{1}$$

$$X_{Oxygen} = 0.9075$$

$$X_{Water\ vapours} = \frac{Pressure\ of\ Water\ vapours}{Total\ pressure\ Mixture} = \frac{0.0925}{1}$$

$$X_{Water\ vapours} = 0.0925$$

$$Total\ mole\ fraction = X_{Oxygen} + X_{Water\ vapours}$$

$$Total\ mole\ fraction = 0.9075 + 0.0925 = 1$$

...i (a) What is Viscosity? Why viscosity decreases with the rise of ...perature?

...SWER:

...COSITY

...decrease in the flow of liquids is due to the internal resistance among ...layers of molecules. Every layer offers some friction to the ...hboring layer which tends to flow over it.

...FINITION

...scosity is the measure of internal resistance of a liquid to flow".

...ASON

...viscosity of liquids decreases with the rise of temperature due to ...ease in kinetic energy of molecules

...vi) (b) Differentiate between any one of the following:

...morphism and polymorphism

...nic solids and covalent solids

...SWER:

...MORPHISM AND POLYMORPHISM

...SWER: (See Question No. 2[vi (ii)], 2023)

...NIC SOLIDS AND COVALENT SOLIDS

IONIC SOLIDS	COVALENT SOLIDS
...nic solids do not conduct ...ectricity.	Covalent solids are poor conductors of heat and electricity
...nic solids possess high melting ...d boiling points.	Covalent solids possess low melting and boiling points
...nic solids are soluble in polar ...vents and insoluble in non-...lar solvents.	Many covalent solids are soluble in water nonpolar liquids but not in water

...ii) State Le-Chatlier principle and discuss its application in the ...thesis of ammonia by Haber's process.

...SWER:

LE-CHATELIER'S PRINCIPLE

In 1884, Le-Chatelier, a French chemist introduced a general rule af long research on how and why the balance of a chemical equilibrium disturbed by some change of conditions, this is known as Le-Chatelier principle.

STATEMENT:

It state that 'If an external stress such as concentration pressure or temperature is applied to a system at equilibrium, the equilibrium is disturbed and tends to shift in a direction to offset of stress imposed'.

OR

When a chemical system at equilibrium is disturbed by the changing concentration, temperature or pressure or catalyst, the position of equilibrium changes so as to minimize the effect of disturbance and establish a new equilibrium.

OR

When a system at equilibrium is disturbed, the equilibrium position shift in the direction which tends to minimize, or counteract, the effect the disturbance.

INDUSTRIAL APPLICATION OF LE-CHATELIER'S PRINCIPLE

HABER'S PROCESS

The synthesis of ammonia by Haber's process is an industrial application of Le-Chatelier's principle.

$$N_{2(g)} + 3H_{2(g)} \leftrightharpoons 2NH_{3(g)} (\Delta H = -ve)$$

The equation tells us that the reaction is exothermic and proceeds with decrease in number of moles of product. To get the maximum promising yield of ammonia following conditions of Le-chatelier's principle should be maintained.

(I) EFFECT OF PRESSURE

In the given equation four moles of gases on the left side and two moles the right side, an increase in the pressure shifts the reaction on the right side and favour the formation of ammonia gas. Since a very high pressure may be dangerous for the process, an optimum pressure should be set. The optimum compromising pressure to a good yield of ammonia is 200-300 atm.

$$N_{2(g)} + 3H_{2(g)} \leftrightharpoons 2NH_{3(g)} (\Delta H = -ve)$$

$$1\,mole + 3\,mole \leftrightharpoons 2\,moles$$

(ii) EFFECT OF TEMPERATURE

As the reaction is exothermic ($\Delta H^o = -ve$), a decrease in temperature shift the reaction to the right side and favour the formation of NH_3. The lowering in temperature do favours the high yield of ammonia but on the other hand it slows down the rate of reaction. The optimum chosen temperature on operational level is 400-500°C.

(iii) EFFECT OF CONCENTRATION CHANGE

Increase in concentration of nitrogen (N_2) or hydrogen (H_2) or decrease the concentration of Ammonia (NH_3) shift the reaction in the forward direction and yield the maximum amount of ammonia.

(iv) ADDITION OF CATALYST

Catalyst speed up the reaction without affecting on equilibrium position. Finely divided Iron is used as a Catalyst in the synthesis of ammonia.

INDUSTRIAL APPLICATION OF LE CHATELIER'S PRINCIPLE

(HABER'S PROCESS)

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Q. (viii) What is Buffer solution? Explain how it resists the change of pH by adding small amount of acid and base.

ANSWER:

BUFFER SOLUTION

It is often necessary to maintain the pH of certain solutions in the laboratory and industrial processes. This can be achieved by the buffer solution.

"A buffer solution is one whose pH is not changed significantly, or even if the small amount of acid or base is added at constant temperature".

OR

A substance that minimizes the change in the acidity or basicity of solution when an acid or base is added to the solution is called a solution.

BUFFER ACTION:

To understand how a buffer works, let us consider the ionization of acid-sodium acetate solution.

$$CH_3COOH_{(l)} \leftrightharpoons CH_3COO^-_{(aq)} + H^+_{(aq)}$$

$$CH_3COONa_{(l)} \leftrightharpoons CH_3COO^-_{(aq)} + Na^+_{(aq)}$$

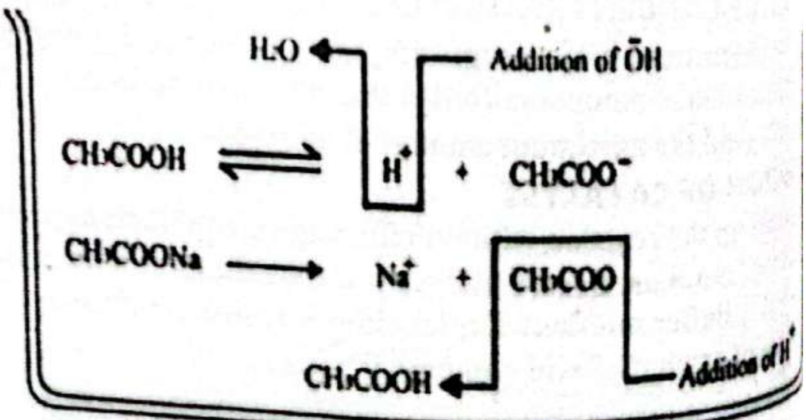
Both acetic acid and sodium acetate provides CH_3COO^- ions, however CH_3COO^- ions that come from CH_3COONa (strong basic salt) are in concentration.

When a small amount of an acid is added to this buffer solution, the H^+ ions of acid combine with excess acetate ions (CH_3COO^-) solution to form acetic acid.

$$CH_3\,COO^-_{(aq)} + H^+_{aq} \leftrightharpoons CH_3\,COOH_l$$

Thus a very slight change in the PH is observed and we say that pH remains practically unchanged.

When a small amount of a base is added to this buffer solution, additional OH^- ions combine with H^+ ions of the buffer to form water molecules. As a result the equilibrium shifts to the right to produce H^+ ions till practically all the excess OH^- ions are neutralized and original buffer pH is restored.



(ix) Enlist various factors which influence on the rate of chemical reaction and describe the effect of temperature on reaction rate.

SWER:

FACTORS EFFECTS ON THE RATE OF CHEMICAL REACTION

Concentration of Reactants
(iii) Concentration of Reactants
Nature of Reactants
(v) Radiations
Surface Area
(vii) Catalyst
Temperature

EFFECT OF TEMPERATURE

The reaction rate is approximately double with the rise of temperature increases the kinetic energy of reacting molecules which enhances the frequency of collision. Although the reactant molecules collide with one another billions times per second but every collision does not lead to a reaction. Only those molecules would react to form products which have energy equal to or greater than certain critical energy known as activation energy.

It is observed that in endothermic reactions, the rate of reaction is doubled when the temperature increases by 10 C or 10K. When the temperature increases by 10 C or 10K the fraction of molecules possessing threshold or activation energy becomes double due to this reason the number of effective collisions is also doubled, therefore the rate of reaction is doubled.

In exothermic reactions, the rate of reaction is decreased with an increase the temperature. Certain reactions like biological reactions that are catalyzed by enzymes may be slowed with an increase the temperature.

2 (x) The reaction $2NO + Cl_2 \rightarrow 2NOCl$ was studied at 25°C. the following results were obtained.

Experiment No	Initial concentration ($mole/dm^3$)		Initial rate ($mole/dm^3.s$)
	NO	Cl_2	
1	0.1	0.1	2.52×10^{-3}
2	0.1	0.2	5.04×10^{-3}
3	0.2	0.1	10.08×10^{-3}

Determine the rate law and order of reaction.

ANSWER: **(See Question No. 2(x), 2024)**

2. (xi) (a) How is a true solution differentiate from suspension.

ANSWER:

SOLUTION IS DIFFERENT FROM SUSPENSION

1. Physical Appearance True Solution is transparent while the suspension is Opaque.
2. The Particle Size in True Solution is Less than 1 nm while in suspension it is above 1000nm.

3. True Solutions cannot be seen by the naked eye while Suspension easily visible by the naked eye.

4. Particles of True Solution do not settle down while Particles of Suspension settled down to their own.

2. (xi) (b) A solution is prepared by dissolving 45g glucose in 72g water, determine mole fraction of glucose and water in the solution.

ANSWER:

DATA:

GIVEN:

$$Mass\ of\ Glucose(C_6H_{12}O_6) = 45g$$
$$Molecular\ mass\ of\ C_6H_{12}O_6 = 72 + 12 + 96 = 180amu$$
$$Mass\ of\ Water(H_2O) = 72g =$$
$$Mass\ of\ Water(H_2O) = 2 + 16 = 18\ amu$$

REQUIRED:

$$Mole\ fraction\ of\ glucose = X_1$$
$$Mole\ fraction\ of\ water = X_2$$

SOLUTION:

$$\because No.of\ moles = \frac{Given\ mass}{Molecular\ Mass}$$

$$No.of\ Mole\ of\ Glucose = \frac{45}{180}$$
$$\therefore n_1 = 0.25$$

$$No.of\ Mole\ of\ Water = $$
$$n_2 = 4$$

$$\because Mole\ fraction = X = \frac{Moles\ of\ substance}{Total\ moles\ of\ Solution}$$

$$X_1 = \frac{n_1}{n_1 + n_2}$$
$$X_1 = \frac{0.25}{0.25 + 4} = \frac{0.25}{4.25}$$
$$X_1 = 0.0588$$

$$X_2 = \frac{n_2}{n_1 + n_2}$$
$$X_2 = \frac{4}{0.25 + 4} = \frac{4}{4.25}$$
$$X_2 = 0.0.941$$

2. (xii) State Raoult's law and derive its mathematical expression in the forms.

ANSWER: (See Question No. 2(xiv), 2023)

2. (xiii) State and explain First Law of thermodynamics. Derive pressure volume work of a system.

ANSWER:

FIRST LAW OF THERMODYNAMICS

In 1847 Helmholtz stated that "energy can neither be created nor destroyed but it may be changed from one form to another". After three years (In 1850) Rudolf Clausius published his statement about the con

...ges in the cyclic process of the thermodynamic system. This ...ent is called the First law of thermodynamics.

...TEMENT:

"...y physical or chemical process, the net energy of a system and its ...ndings must remain constant".

OR

"...stem can not create or destroy energy, however, it can exchange ...y with its surrounding in the form of heat and work".

OR

"...otal energy of a system and its surroundings remains constant, but it ...ssible it converted from one form to another form.

...LANATION:

...velop the mathematical approach to the law let us consider a system ...initial state having internal energy E_1. The system absorbs some ...y "q" from the surroundings while "W" be the work done on the ...m, and then the internal energy increases to E_2. So according to the ...law of thermodynamics, the internal energy changes $(E_2 - E_1)$ may ...rmulated as

$$E_2 - E_1 = q + W$$

$$\Delta E = q + W --- (i)$$

...re "q" is the heat transfer across the system, "W" is the amount of ...k done and "ΔE" is the change in internal energy which is equal to ...E_1.

...N CONVENTION FOR HEAT AND WORK

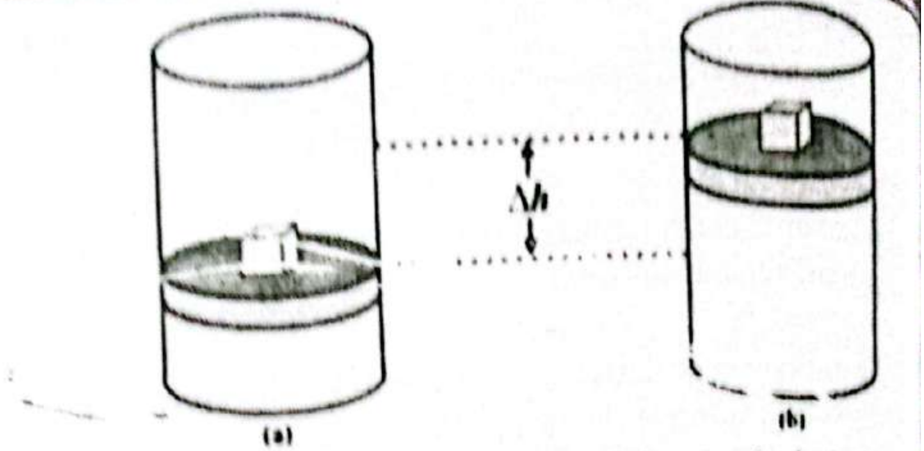
...sign of 'q' is taken as positive if the thermochemical process is ...erned with heat absorption but in the case of heat release, it is taken ...negative sign. The sign of "W" should be taken negatively for ...mochemical expansion and positive if it undergoes compression. ...ation (i) may take the form $\Delta E = q$ if the heat is transferred to the ...m without work is doing but if work is done on the system without ...transfer it may change to $\Delta E = W$.

...SURE-VOLUME WORK:

...hanical work due to expanding or squeezing of a system at constant ...nal pressure is known as pressure-volume work.

...IVATION:

...sider a gas confined in a cylinder filled by a negligible weight, fraction ...nd moveable piston of cross-section area 'A' at a constant external ...ure. If the force 'F' exerted by the gas on the inner wall of the piston ...ater than the external pressure, the piston moves upward from ...l height h_1 to final height h_2 and hence does some work.



$$Work = Force \times Distance(change\ in\ height)$$
$$W = F \times (h_2 - h_1)$$

In this case, the force 'F' is taken as negative because it is exerted by gas on the inner wall of the piston and external pressure opposes the expansion of gas, therefore

$$W = -F \times \Delta h$$

Since pressure is the force exerted on the unit area ($P = \frac{F}{A}$) and the force is ($F = PA$), the work may be written as

$$W = -(PA) \times \Delta h$$
$$W = -P \times \Delta Ah$$

But ΔAh represents volume change (ΔV) of the system so work done in thermodynamics system is deduced as

$$W = -P\Delta V$$

2. (xiv) Calculate the standard enthalpy of formation of carbon disulphide from the given data.

$C + 2S \longrightarrow CS_2$ $(\Delta Hf = ?)$
$C + O_2 \longrightarrow CO_2$ $(\Delta H = -393.5\ KJ/mol)$
$S + O_2 \longrightarrow SO_2$ $(\Delta H = -296.8\ KJ/mol)$
$CS_2 + 3O_2 \longrightarrow CO_2 + 2SO_2$ $(\Delta H = -1075\ KJ/mol)$

ANSWER:

$C + O_2 \longrightarrow CO_2$ $(\Delta H = -393.5\ KJ/mol)$
$S + O_2 \longrightarrow SO_2$ $(\Delta H = -296.8\ KJ/mol)$
$CS_2 + 3O_2 \longrightarrow CO_2 + 2SO_2$ $(\Delta H = -1075\ KJ/mol)$

Equation (*ii*) Multiply by '2'

$2S + 2O_2 \longrightarrow 2SO_2$ $(\Delta H = -593.6\ KJ/mol)$

Adding equation (*i*) and (*iv*)

$C + O_2 \longrightarrow CO_2$ $(\Delta H = -393.5\ KJ/mol)$
$2S + 2O_2 \longrightarrow 2SO_2$ $(\Delta H = -593.6\ KJ/mol)$

$C + 3O_2 + 2S \longrightarrow CO_2 + 2SO_2$ $(\Delta H = -987.1\ KJ/mol)$

Subtracting equation (*iii*) from above equation

$+ 3O_2 + 2S \longrightarrow CO_2 + 2SO_2 \qquad (\Delta H = -987.1\ KJ/mol)$

$CS_2 + 3O_2 \longrightarrow CO_2 + 2SO_2 \qquad (\Delta H = -1075\ KJ/mol)$

$\quad - \qquad\quad - \qquad - \qquad\qquad +$

$C + 2S \longrightarrow CS_2 \qquad (\Delta H = 87.9\ KJ/mol)$

ECTION 'C' (DESCRIPTIVE-ANSWER QUESTIONS)

(Marks: 10x2=20)

ote: Attempt any three (2) questions from this ction. All questions carry equal marks:

Q5 (a) What is an Ideal gas? What are the causes of deviation of real gas m ideal behavior? Explain these deviations at low temperature and h pressure.

SWER:

EAL GAS

"A hypothetical gas which obeys all gas laws (Boyle's, Charles's, ogadro's law,) and kinetic molecular theory at all conditions is called al gas. This gas is also called perfect gas."

USES OF DEVIATION OF REAL GASES FROM IDEAL BEHAVIOR

analyze why real gases deviate from ideal behaviour, we should know e basics on which the ideal gas equation was formulated. The ideal gas ation was obtained from certain assumptions of kinetic molecular ory; two main assumptions are given below.

) The actual volume of the gas molecules is negligibly small as compared the total space of the container.

is assumption remains valid at a low pressure where the volume of gas uch larger due to large intermolecular spaces, but at high-pressure is in the compared state and the volume of gas molecules becomes ificant as compared to the volume of gas enclosed in a container.

) Gas molecules have neither attraction nor repulsion forces.

s assumption is valid at low pressure and high temperature at which ecules tend to be far apart because these forces diminish rapidly as the nce between molecules increases. But at high pressure and low-perature intermolecular forces becomes significant because molecules to be closer together.

(b) Derive an expression for the radius of hydrogen atom in the nth t by using Bohr model.

WER: (See Question No. 3(a), 2023)

(a) Write down the postulates of valence shell electron pair repulsion ry (VSEPR) and predict the shape of the following molecules on the s of VSEPR theory.

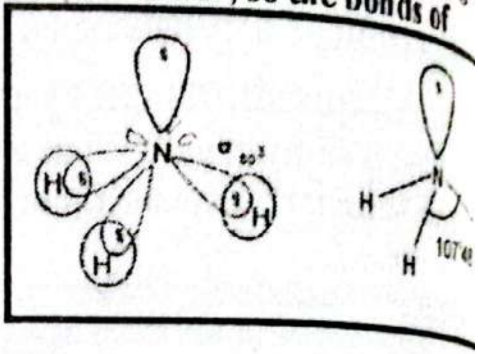
H_3 $\qquad$ * CO_2

WER: (See Question No. 4(a), 2024)

SHAPE OF AMMONIA (NH_3):

ELECTRON PAIR REPULSION MODEL

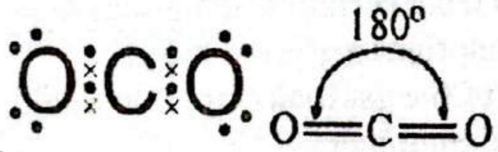
The Lewis structure of NH_3 shows that the central atom of N has three bonding electron and one lone pair of electrons. According to VSEPR theory a lone pair of electron exerts greater repulsion on the bonding electron pairs then bonding pairs do on each other, so the bonds of NH_3 molecule are pushed slightly closer. The bond angle between H-N-H is 107.3° instead of 109.5° predicted form tetrahedral. The shape of ammonia is Trigonal pyramidal.



SHAPE OF CARBON DIOXIDE (CO_2):

ELECTRON PAIR REPULSION MODEL

The Lewis structure of CO_2 shows that the central atom of C has four bonding electrons. In CO_2 molecules' bonds are not involved in the repulsion and geometry because these are inactive electron pairs. Since only two active electron pairs are found around the carbon atom. The bond angle between $O-C-O$ is 180° therefore its shape is Linear.



Q.4 (b) For the reaction

$$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}\ (\Delta H = -ve)$$

If there are 5 moles of SO_2, 3 moles of O_2 and 8 moles of SO_3 are present at equilibrium in a $1dm^3$ flask, at 323K temperature, calculate its Kc and Kp.

ANSWER:

DATA:

GIVEN:

Moles of $SO_{2(g)} = 5$ moles

Moles of $O_{2(g)} = 3$ moles

Moles of $SO_{3(g)} = 8$ moles

Concentration of $SO_{2(g)} = [SO_{2(g)}] = \frac{5}{1} = 5 moles.dm^{-3}$

Concentration of $O_{2(g)} = [O_{2(g)}] = \frac{3}{1} = 3 moles.dm^{-3}$

Concentration of $SO_{3(g)} = [SO_{3(g)}] = \frac{8}{1} = 8 moles.dm^{-3}$

REQUIRED:

$K_c = ?$
and $K_p = ?$

SOLUTION:

$$\because K_c = \frac{Concentration\ of\ Product}{Concentration\ of\ Reactant}$$

$$K_c = \frac{[SO_{3(g)}]^2}{[SO_{2(g)}]^2[O_{2(g)}]}$$

$$K_c = \frac{(8moles.dm^{-3})^2}{(5moles.dm^{-3})^2(3moles.dm^{-3})}$$

$$K_c = \frac{64moles^2.dm^{-6}}{(25\ moles^2.dm^{-6})(9moles.dm^{-3})}$$

$$K_c = \frac{64moles^2.dm^{-6}}{225\ moles^3.dm^{-9}}$$

$$K_c = 0.2844\ dm^3mole^{-1}$$

Q.5(a) What are colligative properties of solution explain elevation of boiling point and depression of freezing point.

ANSWER: (See Question No. 2(xi), 2024)

Q.5 (b) Define Primary and secondary cell. Balance any one of the following equations by ion electron method.

$$Fe^{+2} + Cr_2O_7^{-2} + H^+ \longrightarrow Fe^{+3} + Cr^{+3} + H_2O \quad (acidic\ medium)$$

$$MnO_4^- + SO_3^{-2} \longrightarrow Mn^{+2} + SO_4^{-2} \quad (basic\ medium)$$

ANSWER:

PRIMARY CELLS

The primary Cell is the type of battery, in which the redox reaction takes place in only one direction and cannot be reversed. Thus these batteries cannot be recharged or reused. When all amount of electrolytes present in the cell is consumed, they become dead.

SECONDARY CELLS

Secondary cells are the batteries that can be recharged by reversing the electricity flow through the cell by utilizing a source of external power supply after every use. A lead storage battery which is commonly known as a car battery is an example of the secondary battery.

$$Fe^{+2} + Cr_2O_7^{-2} + H^+ \longrightarrow Fe^{+3} + Cr^{+3} + H_2O \quad (acidic\ medium)$$

SOLUTION:

$Fe^{+2} \longrightarrow Fe^{+3}$ *Oxidation +2 to +3*

$Cr_2O_7^{-2} \longrightarrow Cr^{+3}$ *Reduction +6 to +3*

$Fe^{+2} \longrightarrow Fe^{+3}$

$Cr_2O_7^{-2} \longrightarrow Cr^{+3}$

$Fe^{+2} \longrightarrow Fe^{+3}$

$Cr_2O_7^{-2} + 14H^+ \longrightarrow 2Cr^{+3} + 7H_2O$

$Fe^{+2} \longrightarrow Fe^{+3} + e$

$Cr_2O_7^{-2} + 14H^+ + 6e \longrightarrow 2Cr^{+3} + 7H_2O$

$[Fe^{+2} \longrightarrow Fe^{+3} + e] \times 6$

$[Cr_2O_7^{-2} + 14H^+ + 6e \longrightarrow 2Cr^{+3} + 7H_2O] \times 1$

$6Fe^{+2} \longrightarrow 6Fe^{+3} + 6e$

$Cr_2O_7^{-2} + 14H^+ + 6e \longrightarrow 2Cr^{+3} + 7H_2O$

Adding both equations

$6Fe^{+2} \longrightarrow 6Fe^{+3} + \cancel{6e}$

$Cr_2O_7^{-2} + 14H^+ + \cancel{6e} \longrightarrow 2Cr^{+3} + 7H_2O$

$6Fe^{+2}Cr_2O_7^{-2} + 14H^+ \longrightarrow 6Fe^{+3} + 2Cr^{+3} + 7H_2O$

$MnO_4^- + SO_3^{-2} \longrightarrow Mn^{+2} + SO_4^{-2}$ *(basic medium)*

ANSWER: **(See Question No. 5(a), 2023)**

YOUSUIFS MODEL QUESTION PAPER 1

Annual Examination 2023

CHEMISTRY (THEORY)

CLASS XI (SCIENCE)

Total Time: 3 Hours **Total Marks: 85**

SECTION 'A' (COMPULSORY)

MULTIPLE CHOICE QUESTIONS (M.C.Qs)

NOTE: This Section consists of 17 MCQs and all are to be answered. Each MCQ carry 1 marks.

1. Choose the correct answer from the given options:

(i) Which of the following sample of substance contains the same number of atoms as that of 20_g calcium:

* 16_g S * 20_g C * 19_g K ✓ * 24_g Mg

(ii) (ii) The shape of orbital for which l = 0 is

*Spherical ✓ *Dumbbell *Double dumbbell *Complicated

(iii) (iii) The molecule which has zero dipole moment as:

*NH_3 * HCl * H_2O *CCl_4 ✓

The different rate of C_3H_8 and CO_2 are same:
* Both are poly atomic gases
* Both are denser than air
* Both have same molar mass
* Both contains carbon atoms

The boiling points of different liquids may be different at the same external pressure due to:
* Amounts of liquids
* Intermolecular forces✓
* Surface area
* Viscosities

The temperature at which two allotropic forms co-exist in equilibrium is called as:
* Melting temperature
* Fusion temperature
* Transition temperature ✓
* Critical temperature

The equilibrium of which of the following reaction would not be affected by an increase in pressure:
* $PCl_{5(g)} \leftrightharpoons PCl_{3(g)} + Cl_{2(g)}$
* $2NO_{(g)} + Cl_{2(g)} \leftrightharpoons 2NOCl_{(g)}$
* $N_{2(g)} + O_{2(g)} \leftrightharpoons 2NO_{(g)}$✓
* $2SO_{2(g)} + O_{2(g)} \leftrightharpoons 2SO_{3(g)}$

Which of the following statements is not correct about the bases?
* They have bitter taste
* They have high pH value
* They react with acids to form salts
* They turn blue litmus red✓

The overall order of reaction to which the rate law is R = K:
* Zero order✓
* First order
* Second order
* Third order

Only one pair of liquid in the following set does not obey Raoult's law. Identify it:
* Methanol and Ethanol
* Benzene and toluene
* n-Hexane and n-heptane
* Ethanol and Acetone✓

In a thermochemical process, no work is done if the system is kept at:
* Constant temperature
* Constant pressure
* Constant volume ✓
* Constant mass

This statement is not correct for lead storage battery:
* It can be recharged
* It is primary battery✓
* Anode is made up of lead
* cathode is made up of lad oxide

When 4d orbital is filled, the next electron enter into
* 5s
* 5p✓
* 5d
* 6s

3.01×10^{23} molecules of oxygen gas at STP occupy a volume of:
* $22.4\ dm^3$
* $224\ dm^3$
* $11.2\ dm^3$✓
* $2.24\ dm^3$

The maximum number of electrons in a particular sub-energy level

* $2n^2$✓ * n^2 * $(2l+1)$ * $2(2l+1)$

(xvi) In an exothermic reaction increase of temperature favours:

* Forward Reaction *Reverse Reaction✓
*to remain in Equilibrium * Irreversible Reaction

(xvii) The range of wavelength of x-rays lies between;

* $0.1Å$ *to* $10Å$ ✓ * 10Å to 100Å
* 100Å *to* 500Å * 4000Å *to* 7000Å

SECTION 'B' (SHORT-ANSWER QUESTIONS)

(Marks: 4x10=40)

NOTE: Answer any Ten (10) questions from this section. All questions carry equal marks.

Q.2 (i) (a) Define rounding of data. Give various rules of rounding of data.

ANSWER:

ROUNDING OF DATA:

"To reduce a number upto desired significant figures and adjust the last reported digit is known as rounding off data"

RULES OF ROUNDING OF DATA

(i) If digit to be dropped in greater than 5, then add 1 to the digit to be retained.

For example: 5.768 is rounded up to 5.77 if three significations figures needed to be retained.

(ii) If digit to be drop in less than 5 then simply drop it without changing presiding number for example if 5.734 is rounded up to three significant figures, we get 5.73

(iii) If digit to be dropped is exactly 5, there are two conditions:

* if the digit to be retained in even, then just drop the 5.

For example: when 7.865 is rounded up to three figures we get 7.86

* If the digit to be retained is odd, then add 1 to it

For example 23.35 is rounded to 23.4

Q.2 (i) (b) Calculate each of the Mass in gram of 4.5 moles of ethyne (C_2H_2)

ANSWER:

DATA:

GIVEN:

Number of moles $= 4.5\ moles$

Molecular $mass\ of\ C_2H_2 = 12 \times 2 + 1 \times 2 = 26\ amu$

REQUIRED:

$Mass\ of$ Ethyne =?

SOLUTION:

$$\because Number\ of\ moles = \frac{Given\ Mass}{Molecular\ mass}$$

$$Given\ Mass = Number\ of\ moles \times Molecular\ mass$$

$$Mass\ of\ \text{Ethyne} = 4.5 \times 26$$

$$Mass\ of\ \text{Ethyne} = 117g$$

2 (ii) Mass of 49g of solid potassium chlorate ($KClO_3$) on heating decomposes completely to Potassium Chloride (KCl) with the liberation of oxygen gas(O_2)

$$2KClO_{3(s)} \xrightarrow{Heat} 2KCl_{(s)} + 3O_{2(g)}$$

Determine volume of Oxygen gas laboratory at STP

ANSWER:

DATA:

GIVEN:

Mass of $KClO_3 = 49g$

Molecular $mass\ of$ $KClO_3 = 39 + 35.5 + 48 = 122.5\ amu$

REQUIRED:

$Volume\ of\ O_2 = ?$

SOLUTION:

$$\because Number\ of\ moles = \frac{Given\ Mass}{Molecular\ mass}$$

$$Number\ of\ moles\ of\ KClO_3 = \frac{49}{122.5}$$

$$Number\ of\ moles\ of\ KClO_3 = 0.4\ moles$$

According to the given equation

moles of $KClO_3$ gives 3 moles of O_2

moles of $KClO_3$ gives $\frac{3}{2}$ moles of O_2

moles of $KClO_3$ gives $\frac{3}{2} \times 0.4$ moles of O_2

moles of $KClO_3$ gives 0.6 moles of O_2

know that,

mole of O_2 occupies $22.4dm^3$ at STP.

mole of O_2 occupies $0.6 \times 22.4dm^3$ at STP.

mole of O_2 occupies $13.14dm^3$

$14dm^3$ of Oxygen gas was produced in a laboratory when 49g of solid potassium chlorate ($KClO_3$) was decomposed

2 (iii) Give three properties of each ∝, β and γ rays.

ANSWER:

PROPERTIES OF ALPHA RAYS (α):

1. Alpha particles is composed of two proton and two neutron like the composition of Helium nuclei
2. They are slightly deflected towards negative plate in electric or magnetic field.
3. These particles can ionize gases. Alpha rays have maximum ionizing power.

PROPERTIES OF BETA RAYS (β):

1. Beta particle is composed of electrons
2. They are deflected towards positive plate in electric or magnetic field
3. Their ionizing power is less than that of alpha rays. (It is about one hundredth of alpha particles).

PROPERTIES OF GAMMA RAYS (γ):

1. They are electromagnetic radiations having very short wavelength, in the range of 10^{-10} m to 10^{-13} m.
2. They do not show deflection in electric or magnetic fields.
3. Their ionizing power is low, (about one hundredth of beta particles)

Q.2 (iv) Oil is insoluble in water but soluble in hexane explain why?

ANSWER: The solubility of a substance depends upon its nature, Oil and hexane both are the same in nature i.e. non-polar while water is a polar substance, that's why oil is insoluble in water.

Q.2 (v) A $500cm^3$ vessel contains H_2 gas at 400 torr pressure and another $1dm^3$ vessel contains O_2 gas at 600 torr pressure. If under the similar condition of temperature these gases are transferred to $2dm^3$ empty vessel, calculate the pressure of the mixture of gases in new vessel.

ANSWER:

DATA:

GIVEN:

$$Volume\ of\ First\ Vessel = V_H = 500cm^3 = \frac{500}{1000} = 0.5\ dm^3$$

$$Initial\ Pressure\ in\ First\ Vessel = P_H = 400\ torr$$

$$Volume\ of\ Second\ Vessel = V_O = 1\ dm^3$$

$$Initial\ Pressure\ in\ Second\ Vessel = P_O = 600\ torr$$

$$Volume\ of\ empty\ Vessel = 2dm^3$$

REQUIRED:

$$Final\ Pressure\ in\ empty\ Vessel = P'_H = ?$$

$$Final\ Pressure\ in\ empty\ Vessel = P'_O = ?$$

$$pressure\ of\ the\ mixture\ of\ gases = ?$$

SOLUTION:

FOR HYDROGEN

$$P_H V_H = P'_H V$$

$$P'_H = \frac{P_H V_H}{V}$$

$$P'_H = \frac{(400)(0.5)}{(2)}$$

$$P'_H = 100\ Torr$$

...OXYGEN

$$P_O V_O = P'_O V$$

$$P'_O = \frac{P_O V_O}{V}$$

$$P'_O = \frac{(600)(1)}{(2)}$$

$$P'_O = 300\ torr$$

...ding to Dalton's law of Partial Pressure

$$\text{Total Pressure} = P = P'_H + P'_O$$

$$\text{Total Pressure} = P = 100 + 300$$

$$\text{Total Pressure} = P = 400\ torr$$

...re of the mixture of gases in new vessel is **400 *torr***

...i) (a) Name three major kinds of intermolecular forces in liquids.
...in intermolecular forces in HCl.

...VER:

...rces of attraction among liquid molecules are collectively called ...olecular forces or secondary forces or more famously as Van der ...orces. These attractive forces are actually physically bonds and are ...-static in nature.

...R KINDS OF INTERMOLECULAR FORCES

...ole-Dipole Interaction
...rogen Bonding
...don Dispersion Forces

...MOLECULAR FORCES IN HCl

...molecules are either polar or non-polar. Polar molecules bear ...positive and partial negative charges due to electronegativity ...nces on their opposite ends (dipoles). For example, HCl is a polar ...le and there is a partial positive charge on the hydrogen atom at ...d and a partial negative charge on the chlorine atom at the other ...hese opposite poles attract each other in liquid states. "The ...static force of attraction between the positive end of one polar ...le and the negative end of other polar molecule are called dipole-...interaction". The strength of these forces depends upon the ...nce in electronegativities of the two atoms in the polar molecule. As ...h comparison their strength is about 1% of a normal covalent bond.

...) (b) Explain the following with scientific reason.
...y the compressibility of solids is nearly zero?

(ii).Why some solids are sublime in nature?

ANSWER:

(i). Why the compressibility of solids is nearly zero?

REASON: According to kinetic theory, particles are tightly bonded together due to strong attractive forces therefore only a very small space is left among them due to this reason compressibility of solids is nearly zero

(ii).Why some solids are sublime in nature?

REASON: According to kinetic theory, the attractive forces in sublime solids are quite weaker as compared with ordinary solids, therefore heating high energy surface particles of those solids directly convert vapours.

Q.2 (vii) (a) Define solubility product.

ANSWER:

SOLUBILITY PRODUCT

The product of ionic concentration when dissolved and undissolved are in equilibrium is called the solubility product and it's represented Ksp.

OR

The product of the concentration of ions in a saturated solution of sparingly soluble salt is called the solubility product and it's repre by Ksp.

Q.2 (vii) (b)Write the solubility product expression of the following

BaF_2 , $Li_2C_2O_4$, $MgCo_3$, Ag_3PO_4

ANSWER:

SALTS	EQUATION	SOLUBILI PRODU
BaF_2	$BaF_2 \rightleftharpoons Ba^{2+} + 2F^-$	$Ksp = [Ba^{2}$
$Li_2C_2O_4$	$Li_2C_2O_4 \rightleftharpoons 2Li^+ + C_2O_4^{2-}$	$Ksp = [Li^+]$
$MgCO_3$	$MgCO_3 \rightleftharpoons Mg^{2+} + CO_3^{2-}$	$Ksp = [Mg^{2}$
Ag_3PO_4	$Ag_3PO_4 \rightleftharpoons 3Ag^+ + PO_4^{-3}$	$Ksp = [Ag^+$

Q.2 (viii) Why the aqueous solution of NH_4Cl is acidic and Na_2C alkaline.

ANSWER:

SOLUTION OF NH_4Cl IS ACIDIC

The aqueous solution of ammonium chloride is formed by the r a strong acid with a weak base; therefore it has a higher conce H^+ ion due to this reason acidity of the solution increased, hen solution of ammonium chloride is acidic.

SOLUTION OF Na_2CO_3 IS ALKALINE

Aqueous solution of Sodium Carbonate is formed by the reaction of a strong base with weak acid; therefore it has a higher concentration of OH^-, due to this basicity of the solution increased, hence the solution of sodium carbonate is basic.

(xi) Enlist various factors which influence on the rate of chemical reaction and describe the effect of temperature on reaction rate.

ANSWER: (See Question No. 2(ix) New Model Paper)

(xii) Write the rate law. Determine the order of reaction using the following data:

$$A + B \rightarrow AB$$

Exp.	[A]	[B]	$\frac{dx}{dt}$
1	0.1M	0.1M	$2 \times 10^{3} MS^{-1}$
2	0.2M	0.1M	$8 \times 10^{3} MS^{-1}$
3	0.1M	0.2M	$4 \times 10^{3} MS^{-1}$

ANSWER:

$$K = \frac{R}{[A][B]}$$

Experiment 1:

$$K = \frac{2.0 \times 10^{-3}}{[0.1][0.1]}$$

$$K = \frac{0.002}{0.01}$$

$$K = 0.2\ dm^3 mole^{-1} s^{-1}$$

Experiment 2:

$$K = \frac{8.0 \times 10^{-3}}{[0.2][0.1]}$$

$$K = \frac{0.008}{0.02}$$

$$K = 0.4\ dm^3 mole^{-1} s^{-1}$$

Experiment 3:

$$K = \frac{4.0 \times 10^{-3}}{[0.1][0.2]}$$

$$K = \frac{0.004}{0.02}$$

$$K = 0.2\ dm^3 mole^{-1} s^{-1}$$

Since the value of K is not remain the same for all experiments. Therefore order of reaction $= 1 + 2 = 3$

(xiii) Define osmosis and osmotic pressure.

ANSWER:

OSMOSIS

The surface of certain materials permits water and other smaller size particles to pass through but does not allow large solute particles. These serve as a semi-permeable membrane for example cellophane, animal bladder etc. Although the passage of solvent particles takes place in both directions of the semi-permeable membrane their movement is faster from dilute solution to concentrated solution. This spontaneous process is commonly known as osmosis.

OR

Osmosis is the process of moving solvent molecules from a low-solute-concentration area to a high-solute-concentration area through a semipermeable membrane. Eventually, an equilibrium between the sides of the semipermeable membrane is formed (equal solute concentration on both sides of the semipermeable membrane). Be the semipermeable barrier only allows solvent molecules to pass th no solute particles can pass through.

OSMOTIC PRESSURE

The osmotic pressure can be defined as "the hydrostatic pressure by the solution which stops the process of osmosis".

OR

The least pressure required to apply to a solution in order to stop of solvent molecules across a semipermeable membrane is known osmotic pressure (osmosis). It is a colligative property that is reg the concentration of solute particles in the solution.

2. (xii) Define Molarity and molality. Which of these depends on temperature?

ANSWER:

MOLARITY

The number of moles of solute dissolved in a litter or one decime (dm3) solvent is called the Molarity or molar concentration of th solution.

OR

The number of moles of a solute presents exactly in one liter or decimeter cube (dm^3) of solution. It is represented by M.

Mathematically

$$\text{Molarity} = \frac{\text{Number of moles of solute}}{\text{volume of solution in litter or } dm^3}$$

MOLALITY

The number of moles of solute dissolves in one kilogram (1000 solvent is called Molality or Molal concentration of a solution.

OR

The number of moles of solute present in exactly one kg (1000 solution is called Molality. It is represented by m.

Mathematically

$$\text{Molality} = \frac{\text{Number of moles of solute}}{\text{mass of solvent in kilogram(Kg}}$$

DEPENDENCY ON TEMPERATURE:

Molarity is temperature-dependent because the volume of th decreases or increases with temperature.

2. (xiii) Explain Exothermic and Endothermic reactions with the energy diagram.

ANSWER:

EXOTHERMIC REACTIONS

Exothermic reactions (Exo; out) are those in which heat is given out making the surroundings warmer. In the course of exothermic reactions, energy stored in the chemical bond of products is less than reactant hence these reactions are energetically favourable and often occur spontaneously but sometimes we need extra energy to get them started. The overall energy released in these reactions is represented by ΔH with a negative sign.

ILLUSTRATION:

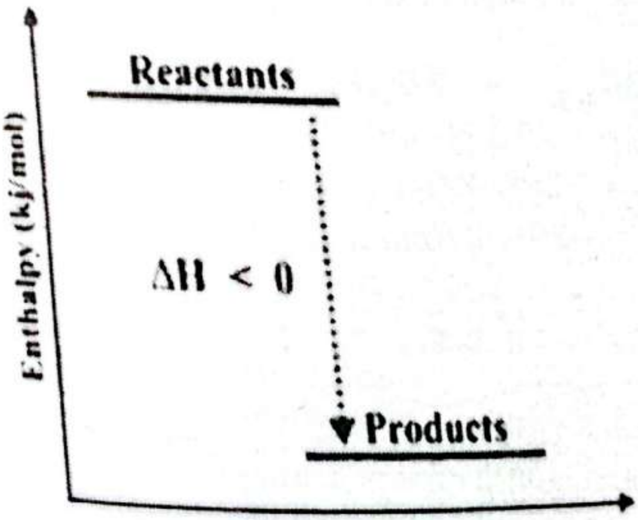
The given diagram, clearly shows that the energy or Enthalpy of reactants is greater than the product. So, Initially, the temperature of the system rises until the highest temperature is reached. When the reaction is completed, then the temperature of the system starts to fall until it reaches room temperature.

EXAMPLE:

All combustion reactions including the burning of fuel and coal, oxidation of Sui gas etc are exothermic reactions and can easily be noted by a rise of the surrounding temperature.

$C + O_2 \rightarrow CO_2 \quad (\Delta H = -ve)$

$$CH_4 + 2O_2 \rightarrow 2CO_2 + 2H_2O \quad (\Delta H = -ve)$$



ENDOTHERMIC REACTIONS

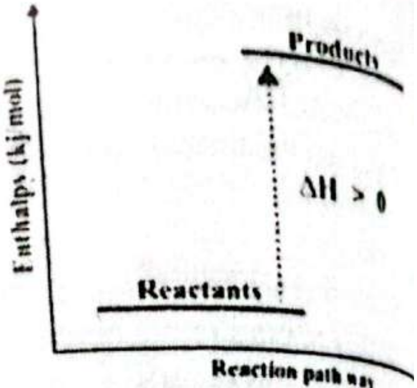
Endothermic reactions (endo; taking in) are those in which heat is put into the system making the surroundings colder. In these reactions, more energy is needed to break the bonds in the reactants than is released when new bonds are formed in the products. Since heat is used for the reaction to occur, these are referred to as non-spontaneous reactions. The net energy absorbed in these reactions is represented by ΔH with a positive sign.

ILLUSTRATION:

The given diagram, clearly shows that the energy or Enthalpy of the product is greater than the reactants. So, Initially, the temperature of system falls until the lowest temperature is reached. When the reaction completed, then the temperature of the system starts to rise until it reaches room temperature.

EXAMPLE:

An appropriate endothermic reaction is a photosynthesis during which plants absorb solar heat through their chlorophyll parts and this heat is utilized for the conversion of CO_2 and H_2O into glucose.



$$2CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2\ (\Delta H = +ve)$$
$$CH_4 \rightarrow C + 2H_2(\Delta H = +ve)$$

2. (xiv) Calculate the standard enthalpy change for each of the follow reaction:

(i) $2H_2S_{(g)} + 3O_{2(g)} \rightarrow 2H_2O_{(l)} + 2SO_{2(g)}$

$\Delta H_f\ of\ H_2S_{(g)} = -20.17J/mol$

$\Delta H_f\ of\ H_2O_{(l)} = -285.8J/mol$

$\Delta H_f\ of\ SO_{2(g)} = -296.8J/mol$

ANSWER:

DATA:

GIVEN:

$\Delta H^o_f\ of\ H_2S_{(g)} = -20.17$

$\Delta H^o_f\ of\ H_2O_{(l)} = -285.8$

$\Delta H^o_f\ of\ SO_{2(g)} = -296.8$

REQUIRED:

Standard Heat of Reaction $= H^o_{reaction} = ?$

SOLUTION:

$$H^o_{reaction} = \{(\sum np\ \Delta H^o_{f\ (products)}) - (\sum nr\ \Delta H^o_{f\ (reactants)})\}$$

$$H^o_{reaction} = [\{2(\Delta H^o_f\ of\ H_2O_{(l)}) + 2(\Delta H^o_f\ of\ SO_{2(g)})\} - \{2(\Delta H^o_f\ of\ H_2S_{(g)})\}]$$

$$H^o_{reaction} = [\{2(-285.8) + 2(-296.8)\} - \{2(-20.17)\}]$$

$$H^o_{reaction} = [\{(-571.6) + (-593.6)\} - (-40.34)\}]$$

$H^o_{reaction} = \{(-1165.2) - (-40.34)\}$

$H^o_{reaction} = (-1165.2 + 40.34)$

$H^o_{reaction} = -1124.86\ KJ/mol.$

(b) The empirical formula of compound is CO_2H. 1.8 g of this compound in gaseous state occupies 448 cm³ at S.T.P find its molecular formula.

ANSWER:

DATA:

GIVEN:

Empirical formula = $CO2H$

Mass of compound = $m = 1.8g$

Empirical Formula Mass = $(12 \times 1) + (16 \times 2) + (1 \times 1)$

$= 12 + 32 + 1 = 45$

Volume = $V = 448cm^3 = \frac{448}{1000} = 0.448dm^3$

Temperature $= T = 273\ K$

Pressure $= P = 1atm$

Gas constant $= 0.0812dm^3 atm\ K - 1mole^{-1}$

REQUIRED:

Molecular formula =?

SOLUTION:

$$\because PV = nRT$$

$$n = \frac{PV}{RT} = \frac{(1)(0.448)}{(0.0812)(273)} = \frac{0.448}{22.167}$$

$$n = 0.020\ mol$$

$$But\ n = \frac{m}{M} \Rightarrow M = \frac{m}{n} = \frac{1.8}{0.020}$$

$$n = 90\ moles$$

$$x = \frac{\text{molecular mass}}{\text{Empirical formula mass}} = \frac{90}{45} = 2$$

Molecular formula=(Empirical Formula)ₓ

$$Molecular\ formula = (CO_2H)_2 = C_2O_4H_2$$

SECTION 'C' (DESCRIPTIVE-ANSWER QUESTIONS)

(Marks: 14x2=28)

Note: Attempt any three (2) questions from this section. All questions carry equal marks:

(a) Write down the postulates of Kinetic molecular theory of gases.

ANSWER:

POSTULATES OF KINETIC MOLECULAR THEORY

The basic postulates of the kinetic theory of gases are as follows.

(i) Gases consist of a large number of tiny particles called molecules. Molecules may be mono-atomic (He, Ne, Ar), diatomic (O_2 ,N_2) or polyatomic (CH_4 , C_4H_{10}).

(ii) Gas molecules are far away from each other and occupy negligible volume as compared to the total volume of the container.

(iii) Gas molecules are in continuous random motion, travelling in a straight line until they collide with each other or with the walls of the container. The average distance covered by the gas between successive collisions is known as the mean free path.

(iv) Give particles undergo elastic collisions (a collision in which no loss or gain of energy takes place).

(v) Gas exert pressure when molecules collide with the walls of the container.

(vi) There is no force of attraction or repulsion found among ideal gas molecules, each molecule acts quite independently.

(vii) The average kinetic energy of gas molecules depends upon absolute temperature. Thus when absolute temperature increases the kinetic energy of the molecules increases.

$$K.E \propto Absolute\ temperature$$

Q.3 (b) State the postulate of Bohr's theory.

ANSWER: (See Question No. 3(a), 2024)

Q.4 (a) What do you mean by Hybridization? Explain sp^3 hybridization in CH_4 molecule and sp^2 hybridization in C_2H_4 molecule.

ANSWER:

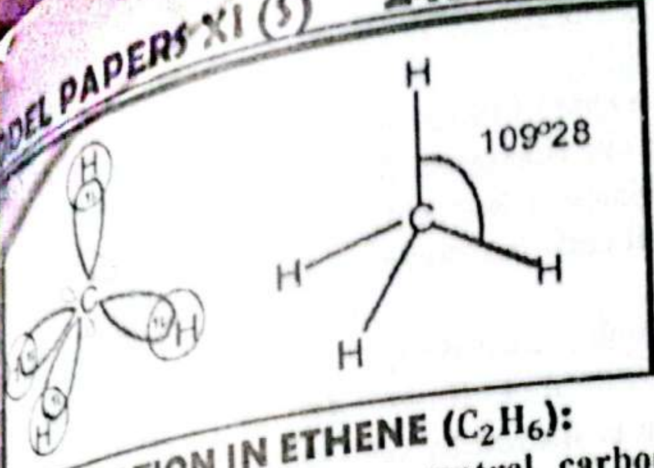
HYBRIDIZATION

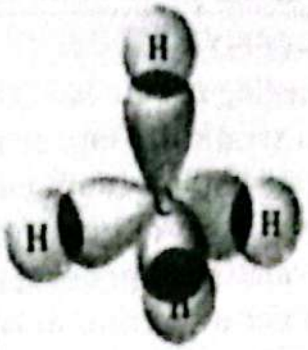
"The mixing of different atomic orbital to produce the same number of equivalent orbital, having same shape and energy is known as hybridization".

sp^3 HYBRIDIZATION IN METHANE (CH_4):

The electronic configuration of carbon is ($1s^2, 2s^2, 2px^1, 2py^1, 2pz^0$) but it is assumed that one electron of 2s orbital get promoted to 2pz orbital to make it tetravalent. C ($z = 6$) $1s^2, 2s^2, 2px^1, 2py^1, 2pz^1$ (tetravalent)

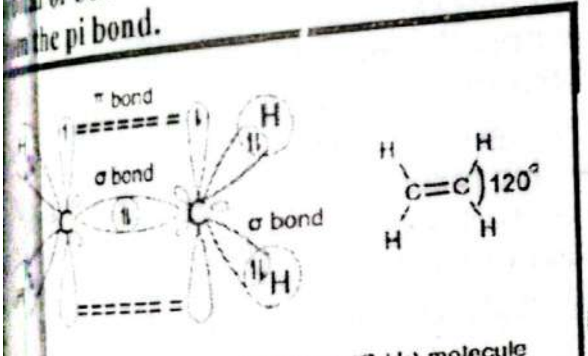
In Methane, the four atomic orbitals of carbon undergo a mixing process to produce four new orbitals of equal energy and shapes, each named as sp^3 hybrid orbital. All four sp^3 hybrid orbitals are then directed towards the corner of a regular tetrahedron at the angle of 109.5°. Since each hybrid orbital has one electron, it overlaps with the 1s orbital of the hydrogen atom. Thus four C-H sigma bonds are formed.



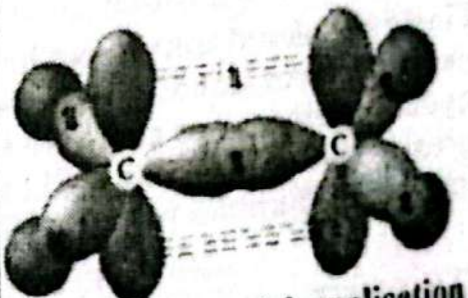


HYBRIDIZATION IN ETHENE (C_2H_6):

...e molecules consist of two central carbon atoms. Each carbon ...sses three sp^2 hybrids orbital and one un-hybrid pz orbital arrange ...angular planar geometry at 120° angle. Two sp^2 orbitals of each ...on overlap with the s orbital of two hydrogen atoms to form two $C-$ ...gma bonds. The third sp^2 orbital of both carbon atoms overlaps ...arly with each other to form a $C-C$ sigma bond. The un-hybrid pz ...tal of both carbon atoms which lie at the right angle overlap sidewise ...m the pi bond.



Planar structure of ehtylene (C_2H_4) molecule



...4 (b) State Le-Chatelier's principle. Explain the industrial application of Le-Chatelier's principle using Haber's process.

ANSWER: (See Question No. 2(vii) New Model Paper)

...5(a) Enlist various factors which influence on the rate of chemical reactions and describe the effect of concentration and surface area of reactants on the reaction rate.

ANSWER:

LIST OF VARIOUS FACTORS

There are following six factors that affect the rate of reaction.

- Concentration of Reactants
- Nature of Reactants
- Temperature
- Effect of Light or radiations
- Surface Area
- Catalyst

CONCENTRATION

According to rate law, the rate of reaction increases with increasing the concentration of one or more reactants. More crowding of reactant, molecules allow more collision to happen.
Thus chances of successful collision increase resulting in an increase rate of formation of products.
Consider a reaction in which A and B react to form a product AB.

$$A + B \rightarrow AB$$

Let's suppose the reaction is obeyed by second order kinetics the rate may be expressed as R = K [A][B]. If the concentration of A is doubled the number of collision among particles of A and B per unit time would also be doubled. Consequently the rate of reaction increases by a factor two fold. Now if we doubled the concentration of both A and B, the collision per unit time has therefore doubly doubled. It increases the rate of reaction four times.

SURFACE AREA

The surface area of a solid is an important factor to enhance the rate of reaction. The greater the surface area the more the possibility of reactant particles colliding with each other and increasing the reaction rate.
Finely powdered substances have a greater surface area as compared to chunks. For example, powdered zinc reacts more vigorously with dilute hydrochloric acid as compared to its chips because powder zinc offers greater surface area for HCl to act upon. This can be demonstrated by intensity of bubbles produced by the liberation of hydrogen gas.

$$Zn_{(s)} + 2HCl_{2(aq)} \rightarrow ZnCl_{2(s)} + H_{2(g)}$$

Q.5 (b) Define Oxidizing agent and reducing agent. Balance any one following equations by ion electron method.

(i) $Cr_2O_7^{-2} + I_2 \longrightarrow Cr^{+3} + IO_3^-$ (acidic medium)

(iv) $Cl_2 + OH \longrightarrow Cl^- + ClO_3^- + H_2O$ (base medium)

ANSWER:

OXIDIZING AGENT

The oxidizing agent is a chemical species that transfers at least one electronegative atom to a chemical species in a chemical reaction. The transferred atom is typically an oxygen atom.

OR

An oxidizing agent is a chemical species that tend to oxidize other substances, i.e. cause an increase in the oxidation state of the substance making it lose electrons.

(i) $Cr_2O_7^{-2} + I_2 \longrightarrow Cr^{+3} + IO_3^-$ (acidic medium)

ANSWER: (See Question No. 3[b (ii)], 2024)

(iv) $Cl_2 + OH \longrightarrow Cl^- + ClO_3^- + H_2O$ (base medium)

SOLUTION:

$\xrightarrow{\quad} Cl^- + ClO_3 + H_2O$

$+OH^-$

Reduction 0 to − 2
Oxidation 0 to + 5

$\xrightarrow{\quad} Cl^-$

$\xrightarrow{\quad} ClO_3^-$

$\xrightarrow{\quad} 2Cl^-$

$Cl_2 + 6OH^- \xrightarrow{\quad} ClO_3 + Cl^- + 3H_2O$

$\xrightarrow{\quad} 2Cl^-$

$+2e$

$+6OH^- \xrightarrow{\quad} ClO_3^- + Cl^- + 3H_2O + 4e$

$+2e \xrightarrow{\quad} 2Cl^-] \times 2$

$+6OH^- \xrightarrow{\quad} 2ClO_3^- + Cl^- + 3H_2O + 4e | \times 1$

$+4e \xrightarrow{\quad} 4Cl^-$

$+6OH^- \xrightarrow{\quad} 2ClO_3^- + Cl^- + 3H_2O + 4e$

...ng both equations

$+4e \xrightarrow{\quad} 4Cl^-$

$+6OH \xrightarrow{\quad} 2ClO_3^- + Cl^- + 3H_2O + 4e$

$+ +6OH^- \xrightarrow{\quad} 5Cl^- + ClO_3^- + 3H_2O$

YOUSUIFS MODEL QUESTION PAPER 2

Annual Examination 2023

CHEMISTRY (THEORY)

CLASS XI (SCIENCE)

...l Time: 3 Hours **Total Marks: 85**

SECTION 'A' (COMPULSORY)

MULTIPLE CHOICE QUESTIONS (M.C.Qs)

...hoose the correct answer from the given options:

...e minimum number of moles are present in:

- ...m^3 of methane gas at STP✓
- 5 dm^3 of helium gas at STP
- ...dm^3 of hydrogen gas at STP
- ...dm^3 of chlorine gas at STP

...1935 A.D James Chadwick was awarded Nobel Prize ...e......

- ...scovered proton
- He discovered neutron✓
- ...iscovered the radius of hydrogen atom
- ...ave the rules of electronic configuration

(iii) The molecule which has zero dipole moment as:
* NH_3 * HCl * H_2O * CCl_4 ✓

(iv) Which one of the following statement is incorrect about the gas molecules?
* They have large spaces
* They possess kinetic en
* Their collision is elastic
* Their molar mass depends upon temperature ✓

(v) Cooking time is reduced in a pressure cooker because:
* Boiling point of water increases ✓
* Boiling point of water decreases
* Vapor pressure of liquid is reduced
* Heat is uniformly distributed

(vi) In NaCl, each Na^+ ion is surrounded by Cl^- ions in the numb
* Four * Three * Six ✓ * Seven

(vii) The term active mass use in law of mass action means:
* No. of mole
* No. of molecules
* mole per dm^3 ✓
* gram per dm^3

(viii) H_2SO_4 is stronger acid than CH_3COOH because:
* It gives two H^+ ion per molecule
* Its boiling point is high
* Its degree of ionization is high ✓
* It is highly corrosive

(ix) The decomposition of H_2O_2 is inhibited by:
* 2 % ethanol * Glycerin ✓ * MnO_2

(x) A colloidal solution of liquid into liquid is known as:
* Gel * Foam * Sol * Emulsion

(xi) Standard enthalpy of formation of all of the following elements at 25°C and 1 atm pressure are zero except:
* $C_{(diamond)}$ * $C_{(graphite)}$ ✓ * O_2 * N_2

(xii) Which of the following half cell reaction show oxidation:
* $F^{+3} \rightarrow F^{+2}$
* $Cl_2 \rightarrow 2Cl^-$
* $SO_2^{-2} \rightarrow SO_4^{-3}$
* $Zn \rightarrow Zn^{+2}$ ✓

(xiii) The molecule which has maximum bond angle:
* CS_2 ✓ * H_2O * NH_3 * BF_3

(xiv) 602.10 has significant figures:
* 3 * 4 * 5 ✓ * 6

(xv) This molecule has the maximum bond angle:
* NH_3 ✓ * CO_2 * SO_2 * H_2O

(xvi) Precipitation occur if the product of ionic concentration is
* greater than K_{sp} ✓
* less than K_{sp}
* equal to K_{sp}
* equal to K_{sp}

(xvii) Cooling appliances like air conditioners and refrigerators are working on the principle of:

* Common ion effect
* Joule-Thomson effect ✓
* Pauli's exclusion principle
* Le-Chatlier's principle

SECTION 'B' (SHORT-ANSWER QUESTIONS)

(Marks: 4x8=32)

NOTE: Answer any Ten (10) questions from this section. All questions carry equal marks.

Q.2 (i) (a) Why the practical yield is often less than theoretical yield?

ANSWER: Usually, the actual yield is lower than the theoretical yield because few reactions truly proceed to completion (i.e., aren't 100% efficient) or because not all of the product in a reaction is recovered.

Q.2 (i) (b) Number of molecules of 126g water

ANSWER:

DATA:

GIVEN:

Given mass of $H_2O = 126\ g$

Molecular mass of $H_2O = 1 \times 2 + 16 \times 1 = 18\ amu$

REQUIRED:

Number of molecules of H_2O =?

SOLUTION:

$$\because \text{Number of moles} = \frac{\text{Given Mass}}{\text{Molecular mass}}$$

$$\text{Number of moles} = \frac{126}{18}$$

$$\text{Number of moles} = 7$$

$$\because \text{Number of molecules} = \text{Number of moles} \times N_A$$

$$\text{Number of molecules of } H_2O = 7 \times 6.02 \times 10^{23}$$

$$\text{Number of molecules of } H_2O = 42.14 \times 10^{23}$$

OR

$$\text{Number of molecules of } H_2O = 4.214 \times 10^{24}\ \text{molecules}$$

Q.2 (ii) Aluminum chloride is used in the manufacturing of rubber. It is produced by allowing Aluminum to react with Cl_2 gas at 650°C.

$$2Al_{(s)} + 3Cl_{2(g)} \longrightarrow 2AlCl_{3(l)}$$

When 160g Aluminum reacts with excess of chlorine, 650g of $AlCl_3$ is produced. What is percentage yield of $AlCl_3$?

ANSWER:

DATA:

GIVEN:

Mass of Aluminum $= 160g$

Atomic mass of $Al = 27 \times 1 = 27\ amu$

REQUIRED:

Percentage yield of $AlCl_3 = ?$

SOLUTION:

$$\because Number\ of\ moles = \frac{Given\ Mass}{Molecular\ mass}$$

$$Number\ of\ moles\ of\ Al = \frac{160}{27}$$

$$Number\ of\ moles\ of\ Al = 5.926\ moles$$

MOLE COMPARISON OF Al AND $AlCl_3$

According to Balanced chemical Equation

2 mole of Al gives = 2 mole of $AlCl_3$

1 mole of Al gives = 1 mole of $AlCl_3$

5.926 moles of Al gives = 5.926 moles of $AlCl_3$

$$\because Theoretical\ yield = Moles\ of\ product \times Molecular\ mass$$

$$Theoretical\ yield = 5.926 \times 133.5$$

$$Theoretical\ yield = 791.121$$

$$\because Percentage\ yield = \frac{Practical\ yield}{Theoretical\ yield} \times 100$$

$$Percentage\ yield = \frac{650}{791.121} \times 100$$

$$Percentage\ yield = 82.161\%$$

Q.2 (iii) What is the shape of orbital for which l = 0 and l = 1.Write electronic configuration of the following species:

*Cu (Z=29) *Ag(Z=47)

ANSWER:

SHAPE OF ORBITAL

VALUE OF AZIMUTHAL QUANTUM NUMBER	SHAPE OF ORBITAL
$l = 0$	orbital has spherical shape
$l = 1$	orbital has dumb-bell-shape

ELECTRONIC CONFIGURATION

ELEMENTS	CONFIGURATION
$Cu\ (Z = 29)$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1, 3d^{10}$
$Ag\ (Z = 47)$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}, 4p^6, 5s^1 \ldots$

Q.2 (iv) Define the following:

(i) Bond Energy (ii) Bond length

ANSWER:

OND ENERGY

...ergy involved for the breaking / formation of 1 mole of particular ...f bonds in the molecule is known as bond energy".

OND LENGTH

...distance between the nuclei of two bonded atoms in a molecule is ...n as bond length".

...) Calculate the volume occupied by 8g of methane gas at 40°C and ...rr pressure?

...WER:

...A:

...EN:

...n mass of methane $= m = 8g$

...ecular mass of $CH_4 = M = 12 \times 1 + 1 \times 4 = 16$

...perature $= 40°C = 40 + 273 = 313K$

...ssure in torr $= P = 842\text{torr}$

...ssure in atm $= P = \frac{1}{760} \times 842 = 1.10789atm$

...s Constant $= R = 0.0821$ atm $dm^3 mole^{-1} K^{-1}$

...QUIRED:

...me of methan $= V = ?$

...UTION:

...rding to Ideal Gas Equation

$$\because PV = nRT$$

$$PV = \left(\frac{m}{M}\right) RT$$

$$V = \frac{mRT}{PM}$$

$$V = \frac{(8)(0.0821)(313)}{(1.10789)(16)}$$

$$V = \frac{205.5784}{17.72624}$$

$$V = 11.597dm^3$$

...he volume occupied by 8g of methane gas at 40°C and 842 torr pressure ...11.597dm^3

...2 (vi) (a) Give reasons for the following:

...) Evaporation is cooling process.

...NSWER: (See Question No. 2[v(i)], 2024)

...i) Mercury has its meniscus upward.

...NSWER: Mercury has more strong cohesive forces than adhesive ...rces. Therefore mercury falls down from the sides attached to the wall ...nd formed a concave meniscus.

Q.2 (vi) (b) Define lattice energy. Explain how it is effected by size and charge of ion.

ANSWER:

LATTICE ENERGY

Lattice energy is effected by size and the charge of ion. In the formation of ionic solids, appositively charged gaseous ions are brought closer to each other and arranged three-dimensionally in a certain pattern. During this process, a high amount of energy is released known as lattice energy. It is the energy associated with electrostatic interaction between the ions in the crystal and may be defined as "the quantity of energy released when one mole of the ionic crystal is formed from the gaseous ions". It is also defined as the energy required to break one mole of crystalline solid into isolated ions in the gas phase. It is expressed in $KJ\ mole^{-1}$.

$$Na_g + Cl_g \rightarrow NaCl_s\ (\Delta H = -787KJ/mol)$$

$$NaCl(s) \rightarrow Na^+_{(g)} + Cl^-_g\ (\Delta H = +787KJ/mol)$$

LATTICE ENERGY IS AFFECTED BY THE SIZE AND THE CHARGE OF ION

The lattice energy decreases with the increase in the size of cation or anions. The reason in both cases is the same. The smaller the size of cation or an ion the closer the packing of appositively charged ions thus require high energy to break the lattice and convert the solid into isolated ions.

Lattice energy is also affected by the charge of ions. The greater the charge of the ion the higher the lattice energy. For example lattice energy of BeF_2 (3505 KJ/mol) is much higher than LiF (1036) because Li possesses + 1 charge in LiF whereas the charge of beryllium is +2.

Q.2 (vii) State law of mass action and derive Kc expression of a reversible reaction.

ANSWER:

LAW OF MASS ACTION

"The rate at which a substance reacts is proportional to its active mass and the rate of a chemical reaction is proportional to the product of active masses of reactants".

OR

"The rate of a chemical reaction is directly proportional to the product of the molar concentrations of the reactants at a constant temperature at any given time".

EXPRESSION OF A GENERAL REVERSIBLE REACTION

Consider, a general reversible reaction in which reacting species are A, B, C and D while a, b, c and d are the number of moles of A, B, C and D.

$$aA + bB \rightleftharpoons cC + dD$$

$$K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

(viii) Describe Lewis theory of acids and bases.

SWER:

IS DEFINATION OF ACIDS AND BASES

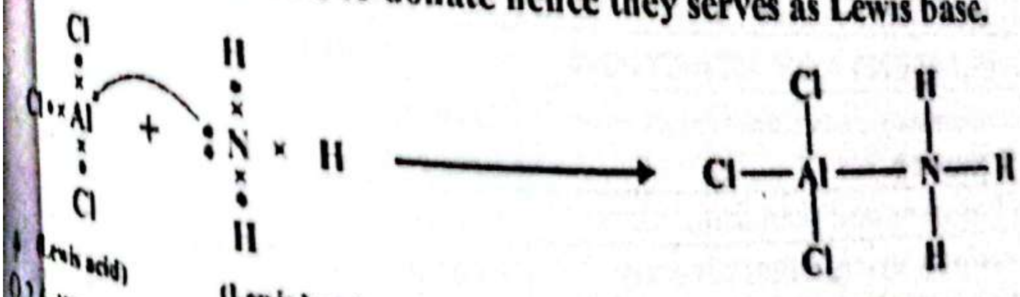
rding to G.N Lewis (1923) "Acid is a substance that accepts an ron pair whereas the base is a substance that can donate electron ". Lewis acid and base is commonly Known as electrophilic (electron-g) and nucleophilic (nucleus-loving) reagents since they have bility to accept and donate electron pair respectively.

Consider Lewis concept, hydrogen ion (H^+) is an acid because it pts electron pair. Similarly hydroxide ion (OH^-) is base since it ate lone pair of electron.

Lewis theory does not explain about the transfer of proton from specie to another however it indicates sharing of electron pair een a donor and accepter reagent. Generally species (compounds or ons) having less than a full octet of electrons behave like Lewis acids all species (compounds or anions), having lone pair of electron behave Lewis base. The product formed in a lewis acid base reaction esses a coordinate covalent bond between lewis acid and base and n as adduct.

$$A + :B \rightarrow A - B$$

ral neutral molecules such as $AlCl_3$, BF_3, $FeCl_3$ etc severs as Lewis id due to electron deficiency on the central atom (fewer than eight nce electrons). On the other hand ammonia (NH_3), Phosphene (PH_3) ter (H_2O) etc although have their complete octet but possess lone pair f electron that is available to donate hence they serves as Lewis base.



(Lewis acid) (Lewis base) (Adduct)

Q.2 (viii) (OR) The hydroxide ion concentration is an antiseptic solution at 25°C is 3.5×10^{-4}M. Calculate its pH.

ANSWER:

DATA:

GIVEN:

Hydroxide ion concentration $= [OH^-] = 3.5 \times 10^{-4}M$

REQUIRED:

$pH = ?$

SOLUTION:

$$\because pOH = -log[OH^-]$$
$$pOH = -log[3.5 \times 10^{-4}]$$
$$pOH = 3.455$$
$$\because pH + pOH = 14$$
$$pH = 14 - pOH$$
$$pH = 14 - 3.455$$
$$pH = 10.544$$

Q.2 (ix) Differentiate between the following:
(i) Homogenous and Heterogeneous catalyst
(ii) Elementary and overall reaction

ANSWER:

(i) Homogenous and Heterogeneous catalyst

HOMOGENEOUS CATALYST	HETEROGENEOUS CATAL
Homogeneous catalysts are catalytic compounds that are in the same phase as the substances which are going into the reaction phase.	Heterogeneous catalysts are catalytic compounds that are different phase from that of the phase of the reaction mixture
Homogeneous catalysts can be found mostly in the liquid phase.	Heterogeneous catalysts can be found in all three phases; solid phase, liquid phase or gas pha
The thermal stability of homogeneous catalysts is poor.	The thermal stability of heterogeneous catalysts is good
Homogeneous catalysts work better in low-temperature conditions (less than 250°C).	Heterogeneous catalysts work better in high-temperature conditions (around 250 to 500

(ii) ELEMENTARY AND OVERALL REACTION

ELEMENTARY REACTION	OVERALL REACTION
elementary reaction Occurs in single step	Overall reaction is the sum of several steps.
It has only one transition state	It has Series of transition state
Products are formed directly from the reactants	Products are not formed direc from the reactants

2. (x) Determine the order of reaction from the data given below:

$$A + B \longrightarrow Product$$

Experiment Number	[A] $mole/dm^3$	[B] $mole/dm^3$	Rate of reaction $mole/dm^3$
1	0.1	0.1	3.0×10^{-3}
2	0.2	0.1	6.0×10^{-3}
3	0.1	0.3	9.0×10^{-3}

TION:

$$K = \frac{R}{[A][B]}$$

ent 1:
$K = \frac{3.0\times10^{-3}}{[0.1][0.1]}$
$K = \frac{0.003}{0.01}$
$3\ dm^3 mole^{-1}s^{-1}$

Experiment 2:
$K = \frac{6.0\times10^{-3}}{[0.2][0.1]}$
$K = \frac{0.006}{0.02}$
$K = 0.3\ dm^3 mole^{-1}s^{-1}$

Experiment 3:
$K = \frac{9.0\times10^{-3}}{[0.1][0.3]}$
$K = \frac{0.009}{0.03}$
$K = 0.3\ dm^3 mole^{-1}s^{-1}$

he value of K remains same for all experiments. Therefore

Order of reaction $= 1 + 1 = 2$

Name various units of concentration and explain mole fraction?

WER:

S OF CONCENTRATION

larity
2 Molality (m)
e Fraction (X)
4. Mass percent
s Per million (ppm)
6. Parts Per Billion (ppb)
s Per Trillion (ppt)

FRACTION (X)

Fraction is used when two or more components are present in the n. It is the ratio of number of moles of one component to that of all nents of solution.

$$= \frac{\text{moles of substance A}}{\text{total moles of solution}} \qquad X_B = \frac{\text{moles of substance B}}{\text{total moles of solution}}$$

m of mole fractions of all the components of a solution is always a

$$X_A + X_B = 1$$

ample if a solution is made up of 1 mole methanol in 4 moles water, le fraction of methanol will be 0.2 and water will be 0.8. Now if we e mole fraction of both methanol and water we get unity.

$$X_{methanol} + X_{water} = 0.2 + 0.8 = 1$$

Give four characteristics to distinguish between ideal and non-ideal n.

WER:

NGUISHED BETWEEN IDEAL AND NON-IDEAL SOLUTION

lution which perfectly obey the Raoult's law or obey the nship mentioned in equation are referred as ideal solutions. er many solutions do not satisfy Raoult' law due to the difference in ractive forces and molecular size of solute and solvent particles. solutions are formed by the change in volume and enthalpy and as non-ideal solution.

There are four distinct characteristics of an ideal solution.
(i) It obeys Raoult law at all temperatures and concentrations. Wh... ideal not obeys.
(ii) The molecular interaction of liquid A–A, B–B, and A–B remain almost the same.
(iii) No heat absorbs or evolves in the formation of the ideal solution = 0)
(iv) No change in the volume takes place in the formation of the ideal solution ($\Delta V = 0$).
Examples of liquid pairs which form ideal solution and mixing are methanol ethanol, benzene-toluene etc.

2. (xiii) Define Thermochemsitry. State and explain Hess's law of Constant heat summation

ANSWER:

THERMOCHEMSITRY

"The branch of chemistry concerned with the study and measurement the heat evolved or absorbed during chemical reactions".

HESS'S LAW OF ENTHALPY SUMMATION

G.H Hess (1840) introduced a law of thermochemistry based on his observations on cyclic thermochemical changes which involves two more different routes. This is known as Hess's law of enthalpy summation.

STATEMENT:

This law states as "If a chemical reaction can be brought about in than one path way, the net enthalpy change is the same provided the initial and final states are the same".

OR

The energy change for any chemical reaction is independent number of steps required to complete the reaction.

OR

The energy change in converting reactants into products is regardless of the pathway taken, provided that the conditions reactants and products are the same.

EXPLANATION OF HESS'S LAW

According to Hess's law reactant converts into a product in a chemical reaction in one or several steps, in both conditions, an equal amount energy is absorbed or evolved. In other words steps or pathways responsible for energy changes.

Consider a chemical reaction in which A is a reactant and the product is Z. The energy changing in this chemical reaction is... are two ways to complete this chemical reaction. One is a direct and the other is an indirect method. The direct method occurs in

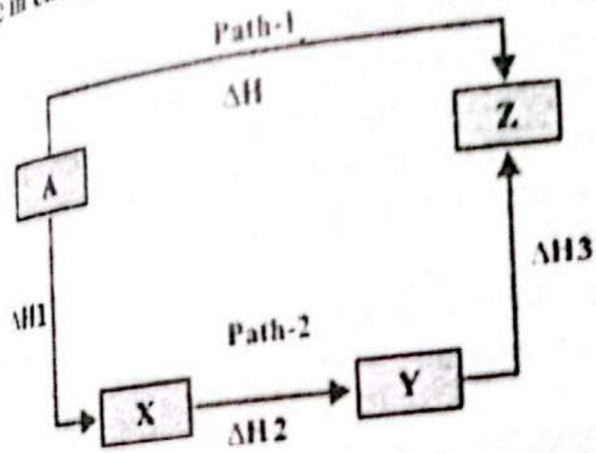
...ctant directly converts into the product. In the indirect method ...A converted into X product then X is converted into Y product ...Y forms the final product Z. The energy changes in the ...ethod are $\Delta H_1, \Delta H_2$ and ΔH_3 respectively.

$$A \xrightarrow{\Delta H} Z$$

$$A \xrightarrow{\Delta H_1} X \xrightarrow{\Delta H_2} Y \xrightarrow{\Delta H_3} Z$$

$$\Delta H_1 + \Delta H_2 + \Delta H_3 = \Delta H$$

...al change in energy is equal to the sum of all energy changes in ...steps.



...) A thermochemical process is carried out at constant pressure of ...m. If it absorbs 15.4KJ energy from the surrounding due to which ...ansion in the volume of $0.47dm^3$ is occurred. Calculate its change ...al energy.

...WER:

...A:

...N:

...ure $= P = 8.52atm$

...gy absorbs $= q_p = 15.4KJ$

...nsion $= \Delta V = 4.7dm^3$

...UIRED:

...ge in Internal Energy $= \Delta E = ?$

...UTION:

...ding to the first law of thermodynamics, process at constant ...ure is

$$\because \Delta E = q_p - P\Delta V$$

$$\Delta E = 15.4 - 8.52 \times 4.7$$

$$\Delta E = 15.4 - 38.775$$

$$\Delta E = -23.375$$

...) Give the postulates of Bohar's atomic theory and derive the ...ula for the radius of nth orbit of hydrogen atom.

ANSWER: (See Question No. 3(a), 2023)

SECTION 'C' (DESCRIPTIVE-ANSWER QUESTI

(Marks: 14x2=28)

Note: Attempt any three (2) questions from this section. All questions carry equal marks:

Q.3 (a) State and explain Dolton's law of partial pressure. Give applications of Dolton's law.

ANSWER:

DALTON'S LAW OF PARTIAL PRESSURE.

In 1801, an English chemist John Dalton gave a law deali the pressure exerted by non-reacting mixture of gases. This law st "The total pressure exerted by a mixture of non-reacting gases in vessel (fixed volume) is always equal to the sum of their individual pressures at a constant temperature".

EXPLANATION

If P_A P_B P_C etc are the individual pressure of gases in the mixture total pressure of the mixture is written as.

$$P_t = P_A + P_B + P_C + \ldots\ldots$$

The pressure exerted by each individual gas in a mixture is called partial pressure of that gas; this is the pressure that individual ga exert if it were alone in the same container at the same temperatu Suppose we have three empty cylinders of equal capacity (1 dm^3) N_2 gases are filled separately in first two cylinder by the O_2 gas a gas are 500 torr and 700 torr respectively. If now these gases are transferred into third empty cylinder under the condition of same temperature; the pressure exerted by the mixture gases is found 1200 torr which is exactly equal to the sum of partial pressure of N_2.

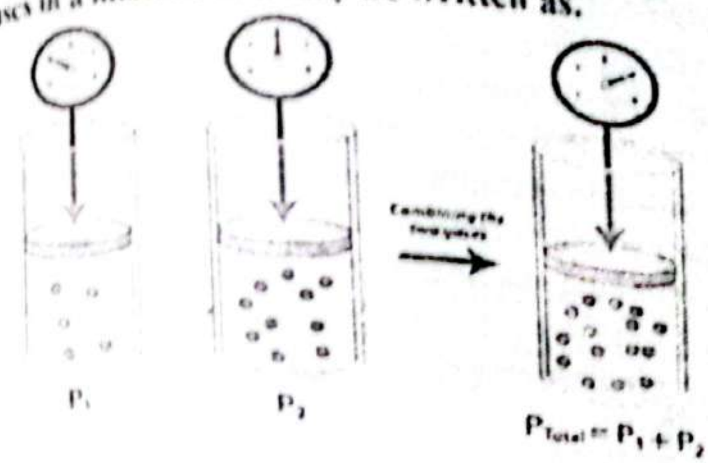
Suppose we have three empty cylinders of equal capacity (1 dm^3) and N_2 gases are filled separately in the first two cylinders by the and N_2 gas are 500 torrs and 700 torrs respectively. If now these transferred into the third empty cylinder under the condition of temperature; the pressure exerted by the mixture gases is found 1200 torr which is exactly equal to the sum of partial pressure of N_2.

According to Dalton's law

$$P_t = P_{O_2} + P_{N_2} = 500 + 700 = 1200 \text{ torr}$$

Since in the mixture, all non-reacting gases behave independently molecules of each gas have equal opportunity to strike with the the cylinder and exert their own pressure without the involvem

...ssure of other gases. Therefore general gas equation can be applied to ...idual gases in a mixture and may be written as.



$$P_A V = n_A RT$$

$$P_A = \frac{RT}{V} n_A \;\ldots\ldots (i)$$

$$P_B V = n_B RT$$

$$P_B = \frac{RT}{V} n_A \;\ldots\ldots (ii)$$

...e R, V and T are constant, therefore

$$P_A \propto n_A \qquad\qquad P_B \propto n_B$$

...s, at constant temperature and volume, the partial pressure of each ...idual gas in a mixture is proportional to its number of moles.

...the mixture of gases, the general gas equation may be written as

$$P_t V = n_t RT$$

$$P_t = \frac{RT}{V} n_t \;\ldots\ldots (iii)$$

...ding equation (i) by (iii), we get

$$\frac{P_A}{P_t} = \frac{n_A}{n_t} \;\ldots\ldots (iv)$$

...arly by dividing equation (ii) with (iii), we get

$$\frac{P_B}{P_t} = \frac{n_B}{n_t} \;\ldots\ldots (v)$$

... mole fraction is the ratio of the number of moles of individual gas ...the total number of moles of all gases present in the mixture.

$$\frac{n_A}{n_t} = X_A \qquad\qquad \frac{n_B}{n_t} = X_B$$

..., equation (iv) and (v) are now reduced as

$$\frac{P_A}{P_t} = X_A \qquad\qquad \frac{P_B}{P_t} = X_B$$

$$P_A = X_A P_t \qquad\qquad P_B = X_B P_t$$

...the equation (vi) and (vii), it is concluded that the partial pressure ...gas of the mixture is equal to the product of the mole fraction of ...s and the total pressure of the mixture. It should be noted that the ...raction of any gas in the mixture is less than one but the sum of the ...action is always equal to one.

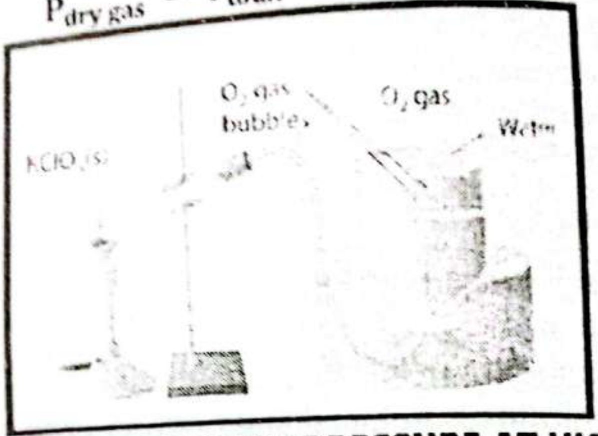
APPLICATIONS OF DALTONS LAW OF PARTIAL PRESSU

(i) PRESSURE OF GASES COLLECTED OVER WATER:

When a gas is collected over water by the downward displacement of water in a gas jar, the pressure of dry gas can be calculated by Dalton's Law. When gas passes through water, it becomes moist and the total pressure will be equal to the sum of the partial pressure of dry gas and water vapours (aqueous tension).

$$P_{total} = P_{dry\ gas} + P_{water\ vapours}$$

$$P_{dry\ gas} = P_{total} - P_{water\ vapours}$$



(ii) MAINTENANCE OF OXYGEN PRESSURE AT HIGH ALTITU

The pressure of air at high altitudes is lower than sea level due to a decrease in the number of molecules of gases, normally, respiration depends upon the difference between the partial pressure of oxygen in air (159 torr) and in the lungs (116 torr). At higher altitudes, the low partial pressure of oxygen causes problems in the process of respirati

(iii) MAINTENANCE OF OXYGEN PRESSURE FOR DEEP SEA DIVERS:

Opposite to altitude, as distance increases downward in the sea, the partial pressure of oxygen increases. At the depth of 40 meters, pres increases to five times. This increased pressure also causes problems respiration. Therefore deep sea divers use the SCUBA (Self Contain Underwater Breathing Apparatus), a breathing tank for respiration. Scuba contains 96% helium gas and 4% oxygen gas.

Q.3 (b) State and illustrate the following rules of electronic configu

a. Pauli's exclusion rule

b. Hund's rule of maximum multiplicity

ANSWER:

PAULI'S EXCLUSION RULE

Pauli's Exclusion Principle was given by Wolfgang Pouli (1925 A.D is based on experimental observation.

...ciple state that, "In an orbital of an atom, no two electrons can ...e same set of four quantum numbers, at least one quantum ... must be different"

...RATION:

...ng to the principle, two electrons in an orbital may have some ...of three quantum numbers (n, l, m) but the value of the fourth ...m number (s) must be different. It means that if one electron of the ...bital has a clockwise spin then the second electron must have an ...kwise spin. Consider two electrons of helium which are lying in s-...of first shell ($1s^2$). The set of four quantum numbers will be ...as:

Helium Atom	Quantum numbers			
	n	l	m	s
First electron	1	0	0	$+\frac{1}{2}$
Second electron	1	0	0	$-\frac{1}{2}$

...auli's principle, it is concluded that an orbital can accommodates ...o electrons and these two electrons must have opposite spins (↑↓).

...'S RULE OF MAXIMUM MULTIPLICITY

...German Physicist Friedrich Hund gave his rule for filling ...s in degenerate orbitals known as the "Hund rule of Maximum ...icity".

...ing to his rule, "In available degenerated orbitals (p, d and f), ...s are distributed in such a way that the maximum number of half-...bitals (single electron in orbitals) are obtained".

...RATION:

...lectrons fill in the orbitals having the same energy, one electron ... each orbital before any orbital has two electrons. Unpair ...s of the same orbit spin in the same direction or parallel and the ...s of the same orbital, spin in opposite directions.

...ve three electrons to fill the $2px$, $2py$, $2pz$ orbitals. We will fill ...ectron in each orbitals $2px^1$, $2py^1$, $2pz^1$ rather than double ...s because $2px^{11}$, $2py^1$, $2pz$.. Unpaired electrons are more stable ...red electrons creating repulsion. The electronic configuration of ...ments considering Hund's rule is given below:

...ent	Atomic number	electronic configuration
...on (C)	6	$1s^{11}\ 2s^{11}\ 2px^1\ 2py^1\ 2pz$
...gen (N)	7	$1s^{11}\ 2s^{11}\ 2px^1\ 2py^1\ 2pz^1$
...en (O)	8	$1s^{11}\ 2s^{11}\ 2px^{11}\ 2py^1\ 2pz^1$
...ine (F)	9	$1s^{11}\ 2s^{11}\ 2px^{11}\ 2py^{11}\ 2pz^1$
...(Ne)	10	$1s^{11}\ 2s^{11}\ 2px^{11}\ 2py^{11}\ 2pz^{11}$

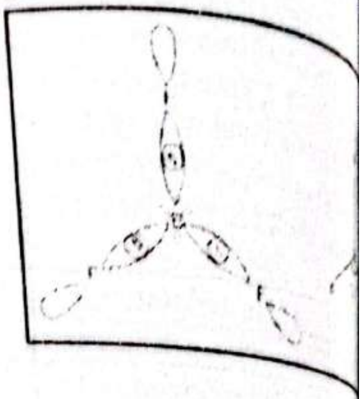
Q.4 (a) Explain the shapes of BF_3 and H_2O on the basis of: Hybrid orbital model and Electron pair repulsion model.

ANSWER:

SHAPE OF BORON TRIFLUORIDE (BF_3):

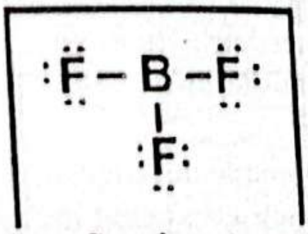
ELECTRON PAIR REPULSION MODEL

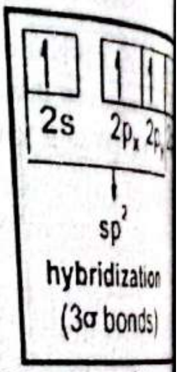
The Lewis structure shows that the central atom of boron has three bonding electron pairs and no un unshared electrons. VSEPR theory says that the three bonding electron pairs adjust as far apart as possible. The shape of BF_3 molecules is Trigonal planar and the angle between F-B-F bonds is 120°.



HYBRID ORBITAL MODE

In the formation of Boron trifluoride, two 2p (p_x, p_y) sub orbital hybrid with one 2s orbital in boron atom to form a new sp^2 hybrid orbital. In boron trifluoride molecule one boron atom attached with three fluorine atoms. The shape of boron trifluoride molecule is Trigonal planar and the angle between F-B-F bonds is 120 °.

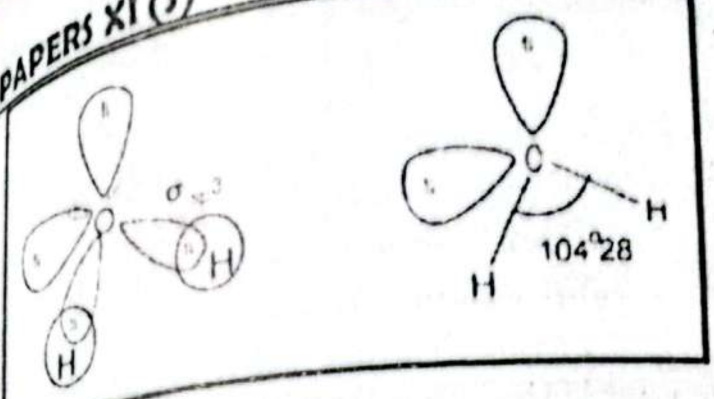




SHAPE OF BORON TRIFLUORIDE (H_2O):

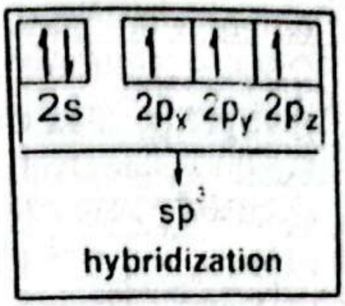
ELECTRON PAIR REPULSION MODEL

The Lewis structure of a water molecule shows that the central oxygen atom is bonded by two hydrogen atoms and has two lone pairs of electrons. In other words, the oxygen atom is surrounded by two bonding electron pairs and two unshared electron pairs are directed to the corners of a tetrahedron but the presence of lone pair of electrons change the shape of the molecules because the lone pair of electron exerts greater repulsion on bonding electron pairs for this reason shape of the water molecule is bent. The angle between bonded H-O-H atoms is 105°.



...ID ORBITAL MODE

...formation of a Water molecule, 2p ...p_y p_z) orbitals hybrid with one 2s orbital ...m a new sp^3 hybrid orbitals. In a water ...ule, one oxygen atom is attached to two ...gen atoms. The shape of the water ...ule is bent and the angle between ...d $H-O-H$ atoms is 105°.



$$H-\ddot{O}-H$$

...) A Solution of $CaCO_3$ is prepared by mixing $200cm^3$ of $2.4 \times$ ...f $Ca(NO_3)_2$ and $300cm^3$ of $4.5 \times 10^{-2}M$ K_2CO_3 will $CaCO_3$ ...itate upon cooling to 25°C if Ksp of $CaCO_3$ at 25°C is 3.8×10^{-9}.

...WER:

...A:

...N:

...ntration of $Ca(NO_3)_2 = 2.4 \times 10^{-4}M$

...me of $Ca(NO_3)_2 = 200cm^3$

...ntration of $K_2CO_3 = 4.5 \times 10^{-2}M$

...me of $K_2CO_3 = 300cm^3$

...f $CaCO_3$ at $25°C = 3.8 \times 10^{-9}$

...Volume $= 200 + 300 = 500cm^3$

...UIRED:

...Product $= Q_{SP} = ?$

...TION:

Concentration of Ion

$$= \left(\frac{\text{Volume of solution}}{\text{Total volume of solution}}\right) \times \text{Concentration of solution}$$

$$[Ca^+] = \left(\frac{200cm^3}{500cm^3}\right) \times 2.4 \times 10^{-4}M$$

$$[Ca^+] = 0.4 \times 2.4 \times 10^{-4}M$$

$$[Ca^{+}] = 9.6 \times 10^{-5} M$$

$$[CO_3^{2-}] = \left(\frac{300cm^3}{500cm^3}\right) \times 4.5 \times 10^{-2} M$$

$$[CO_3^{2-}] = 0.6 \times 4.5 \times 10^{-2} M$$

$$[CO_3^{2-}] = 0.027\ M$$

$$Ionic\ Product = Q_{SP} = [Ca^{+}][CO_3^{2-}]$$

$$Ionic\ Product = Q_{SP} = [9.6 \times 10^{-5}][0.027]$$

$$Ionic\ Product = Q_{SP} = 2.592 \times 10^{-6}$$

Since the value of $Q_{SP} > K_{SP}$ therefore, precipitates of $CaCO_3$ wi formed in the solution

Q.5(a) State Roault's law and drive its mathematical expression.

ANSWER: (See Question No. 2(xiv), 2023)

Q.5 (b) Define Oxidation and reduction. Balance any one of the f equations by ion electron method.

(i) $MnO_4^- + C_2O_4^{-2} + H^+ \longrightarrow Mn^{+2} + CO_2 + H_2O$ (acidic

(ii) $MnO_4^- + SO_3^{-2} \longrightarrow Mn^{+2} + SO_4^{-2}$ (basic medium)

ANSWER:

OXIDATION

According to the modern electronic concept, "Oxidation is the d process in which electrons are lost by an atom or ion".

OR

According to the oxidation number concept "Oxidation is a proc which the oxidation state of an element increases".

REDUCTION

According to the modern electronic concept, "reduction is the d process in which atom or ion gain electron".

OR

According to the oxidation number concept "Reduction is a pro which the oxidation state of an element decreases".

(i) $MnO_4^- + C_2O_4^{-2} + H^+ \longrightarrow Mn^{+2} + CO_2 + H_2O$ (acidic m

SOLUTION:

$$MnO_4^- + C_2O_4^{-2} + H^+ \longrightarrow Mn^{+2} + CO_2 + H$$

Reduction +7

Oxidation +3

$$MnO_4^- \longrightarrow Mn^{+2}$$

$$C_2O_4^{-2} \longrightarrow CO_2$$

$$MnO_4^- + 8H^+ \longrightarrow Mn^{+2} + 4H_2O$$

$$2Cl^- \longrightarrow Cl_2$$

$$MnO_4^- \longrightarrow Mn^{+2}$$

$$C_2O_4^{-2} \longrightarrow 2CO_2$$

$$MnO_4^- + 8H^+ \longrightarrow Mn^{+2} + 4H_2O$$

$$C_2O_4^{-2} \longrightarrow 2CO_2$$

$$MnO_4^- + 8H^+ + 5e \longrightarrow Mn^{+2} + 4H_2O$$

$$C_2O_4^{-2} \longrightarrow 2CO_2 + 2e$$

$$MnO_4^- + 8H^+ + 5e \longrightarrow Mn^{+2} + 4H_2O] \times 2$$

$$C_2O_4^{-2} \longrightarrow 2CO_2 + 2e] \times 5$$

$$2MnO_4^- + 16H^+ + 10e \longrightarrow 2Mn^{+2} + 8H_2O$$

$$5C_2O_4^{-2} \longrightarrow 10CO_2 + 10e$$

Adding both equations

$$2MnO_4^- + 16H^+ + 10e \longrightarrow 2Mn^{+2} + 8H_2O$$

$$5C_2O_4^{-2} \longrightarrow 10CO_2 + 10e$$

$$2MnO_4^- + 16H^+ + 5C_2O_4^{-2} \longrightarrow 2Mn^{+2} + 10CO_2 + 8H_2O$$

(ii) $MnO_4^- + SO_3^{-2} \longrightarrow Mn^{+2} + SO_4^{-2}$ (basic medium)

ANSWER: (See Question No. 5(a), 2023)